

SPCA105N

POSTGRADUATE COURSE

Master of Computer Applications

FIRST YEAR - FIRST SEMESTER

PAPER - V

PRACTICAL - II

RDBMS LAB



**INSTITUTE OF DISTANCE EDUCATION
UNIVERSITY OF MADRAS**

WELCOME

Warm Greetings.

It is with a great pleasure to welcome you as a student of Institute of Distance Education, University of Madras. It is a proud moment for the Institute of Distance education as you are entering into a cafeteria system of learning process as envisaged by the University Grants Commission. Yes, we have framed and introduced as per AICTE Choice Based Credit System(CBCS) in Semester pattern from the academic year 2020-21. You are free to choose courses, as per the Regulations, to attain the target of total number of credits set for each course and also each degree programme. What is a credit? To earn one credit in a semester you have to spend 30 hours of learning process. Each course has a weightage in terms of credits. Credits are assigned by taking into account of its level of subject content. For instance, if one particular course or paper has 4 credits then you have to spend 120 hours of self-learning in a semester. You are advised to plan the strategy to devote hours of self-study in the learning process. You will be assessed periodically by means of tests, assignments and quizzes either in class room or laboratory or field work. In the case of PG (UG), Continuous Internal Assessment for 20(25) percentage and End Semester University Examination for 80 (75) percentage of the maximum score for a course / paper. The theory paper in the end semester examination will bring out your various skills: namely basic knowledge about subject, memory recall, application, analysis, comprehension and descriptive writing. We will always have in mind while training you in conducting experiments, analyzing the performance during laboratory work, and observing the outcomes to bring out the truth from the experiment, and we measure these skills in the end semester examination. You will be guided by well experienced faculty.

I invite you to join the CBCS in Semester System to gain rich knowledge leisurely at your will and wish. Choose the right courses at right times so as to erect your flag of success. We always encourage and enlighten to excel and empower. We are the cross bearers to make you a torch bearer to have a bright future.

With best wishes from mind and heart,

DIRECTOR

COORDINATION & EDITING

Dr. S. Sasikala

Assistant Professor in Computer Science
Institute of Distance Education
University of Madras
Chepauk Chennai - 600 005.

M.C.A.
FIRST YEAR - FIRST SEMESTER
PAPER - V
PRACTICAL - II
RDBMS LAB
SYLLABUS

Objective of the course: This course gives training in design and implementation of data bases for the selected problems. Students are advised to use the concepts like Data Normalization, Link between table by means of foreign keys and other relevant data base concepts for developing databases for the following problems. The implementation of each problem should have necessary input screen Menu-driven query processing and pleasing reports. The choice of RDBMS is left to the students. Necessary validations must be done after developing database.

Course Outcomes: After successful completion of this course, the students should be able to understand and design an application for library management system, student marksheet, bank transaction, quiz using controls and fields.

1. Building Simple Applications.
2. Working with Intrinsic Controls and ActiveX Controls.
3. Application with multiple forms.
4. Application with Dialogs.
5. Application with Menus.
6. Application using Data Controls.
7. Application using Common Dialogs.
8. Drag and Drop Events.
9. Database Management.
10. Creating ActiveX Controls.
11. Library Management System.
12. Students Marksheet Processing.
13. Bank Transactions.
14. Personal information system.
15. Question Database and conducting Quiz.

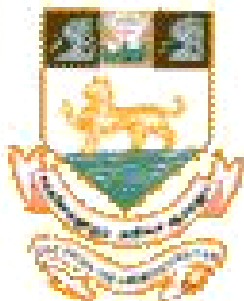
M.C.A.
FIRST YEAR - FIRST SEMESTER
PAPER - V
PRACTICAL - II
RDBMS LAB
SCHEME OF LESSONS

Sl. No.	Title	Page
1.	Students Mark Sheet Processing	26
2.	Electricity Bill Processing	38
3.	Gas Booking and Delivery System	48
4.	Bank Transactions (SB)	57
5.	Inventory System	70
6.	Purchase Order Processing	74
7.	Pay Roll Processing	82
8.	Question Database and Conducting Quiz	96
9.	Library Information Processing	110
10.	Telephone Directory Maintenance	130

INSTITUTE OF DISTANCE EDUCATION UNIVERSITY OF MADRAS

CHENNAI - 600 005

RECORD OF PRACTICALS



M.C.A. (First Year)

20____ - 20____

Practical - II

RDBMS LAB

NAME :

ENROLMENT NUMBER :

GROUP NO. :

INSTITUTE OF DISTANCE EDUCATION UNIVERSITY OF MADRAS

CHENNAI - 600 005.

Certified that this is the Bonafide Record of work done by _____
with Enrolment Number _____ of First Year M.C.A. Degree Course
in the Institute of Distance Education, University of Madras during the year _____
in respect of **RDBMS LAB**

Date:

Co-ordinator

Submitted for First Year M.C.A. Degree Course Practical Examination held on
_____ at _____

Date:

Examiners

1. **Name:**

Signature:

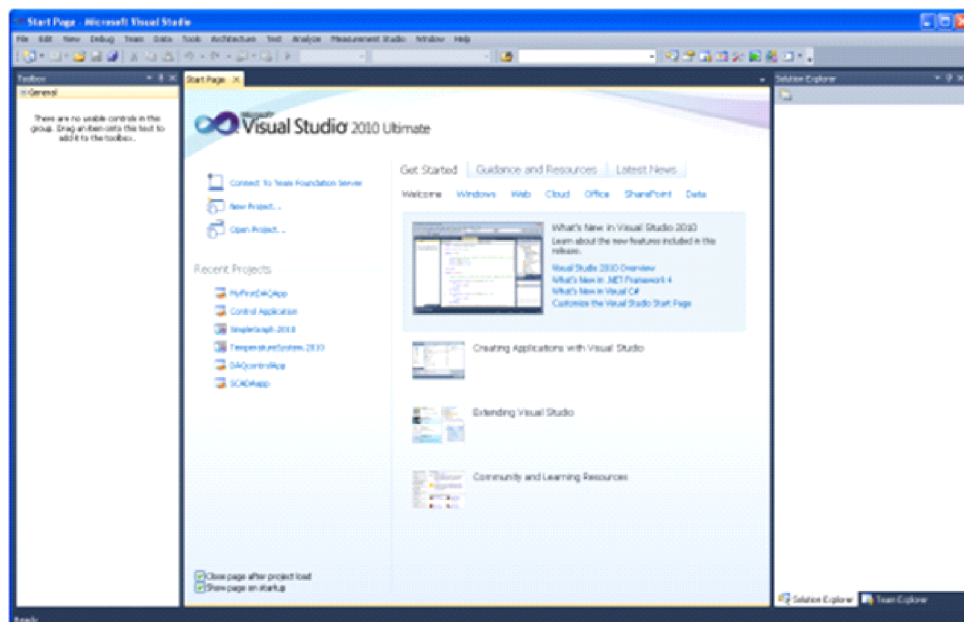
2. **Name:**

Signature:

INTRODUCTION

Getting started to visual studio 2010

The Visual Studio product family shares a single integrated development environment (IDE) that is composed of several elements: the Menu bar, Standard toolbar, various tool windows docked or auto-hidden on the left, bottom, and right sides, as well as the editor space. The tool windows, menus, and toolbars available depend on the type of project or file you are working in. Below we see the Visual Studio IDE (Integrated Development Environment):



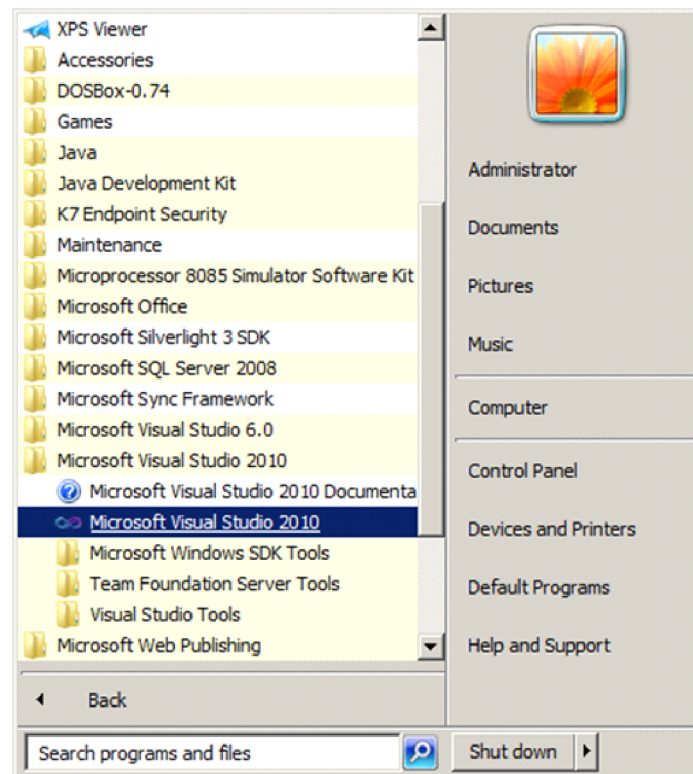
Logon to visual studio 2010 in windows 2007 OS

Step1 → Start → All programs → select the **Microsoft Visual Studio 2010**

Or

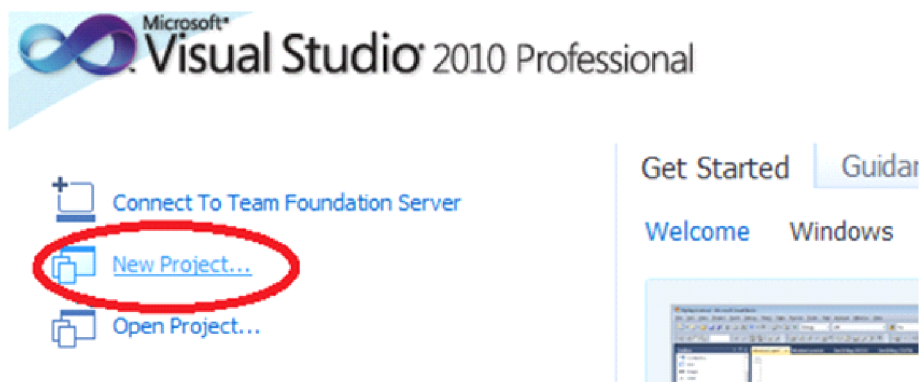
Logon to visual studio 2010 in windows 2010 OS

Start → “ask me anything” option and type “Microsoft Visual Studio 2010”



New Project

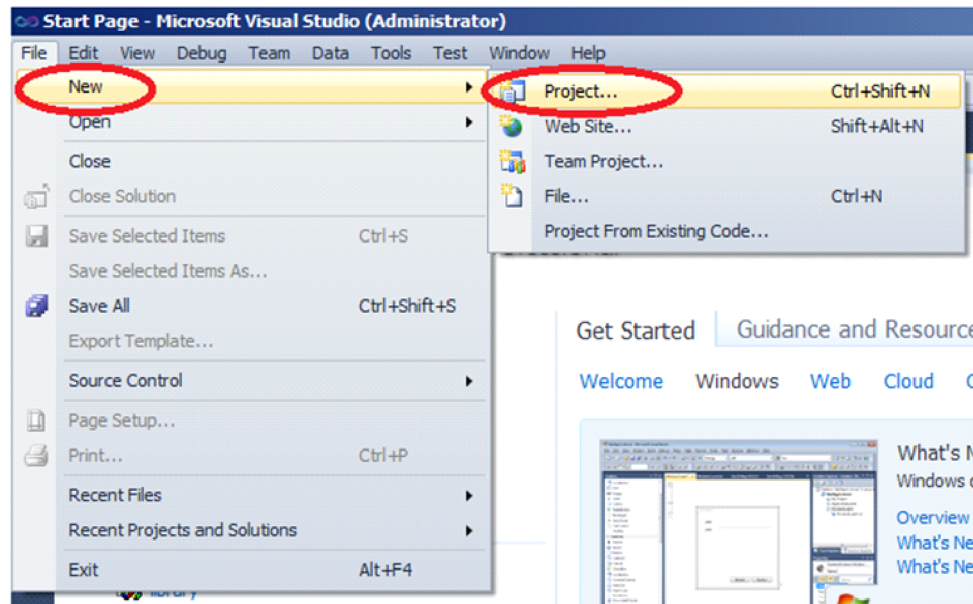
The first thing you do when you want to create a new application is to create a new project. This can be done from the Start Page:



Or from the **file Menu**

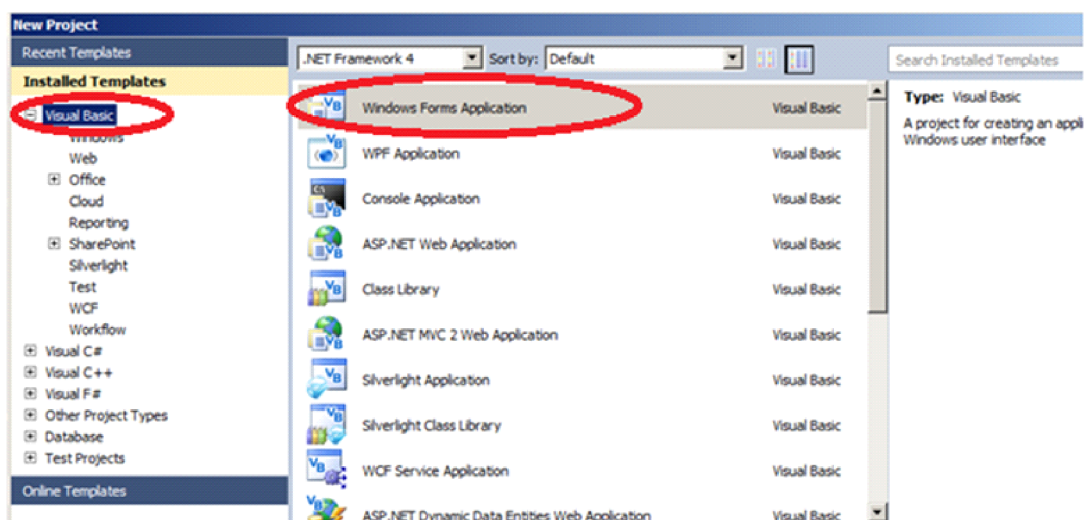
New → project

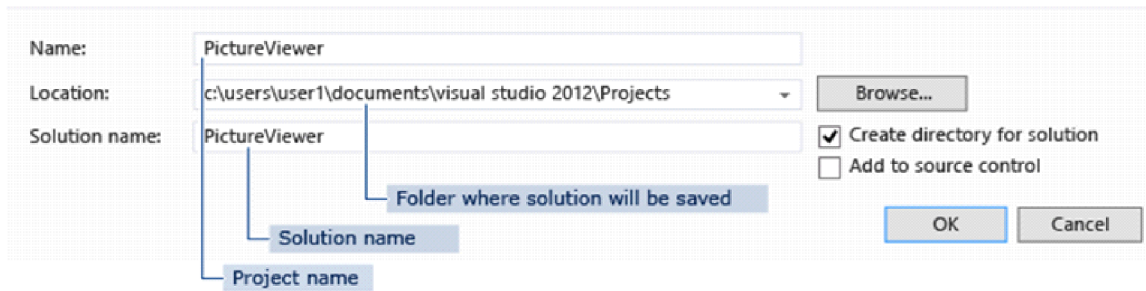
If you want shortcut option in keyboard → CTRL+SHIFT+N and appeared one dialog box



Step 2:

The New Project dialog box opens, as shown on the following page





The New Project dialog box provides access to the major project types available for writing Windows and Web applications.

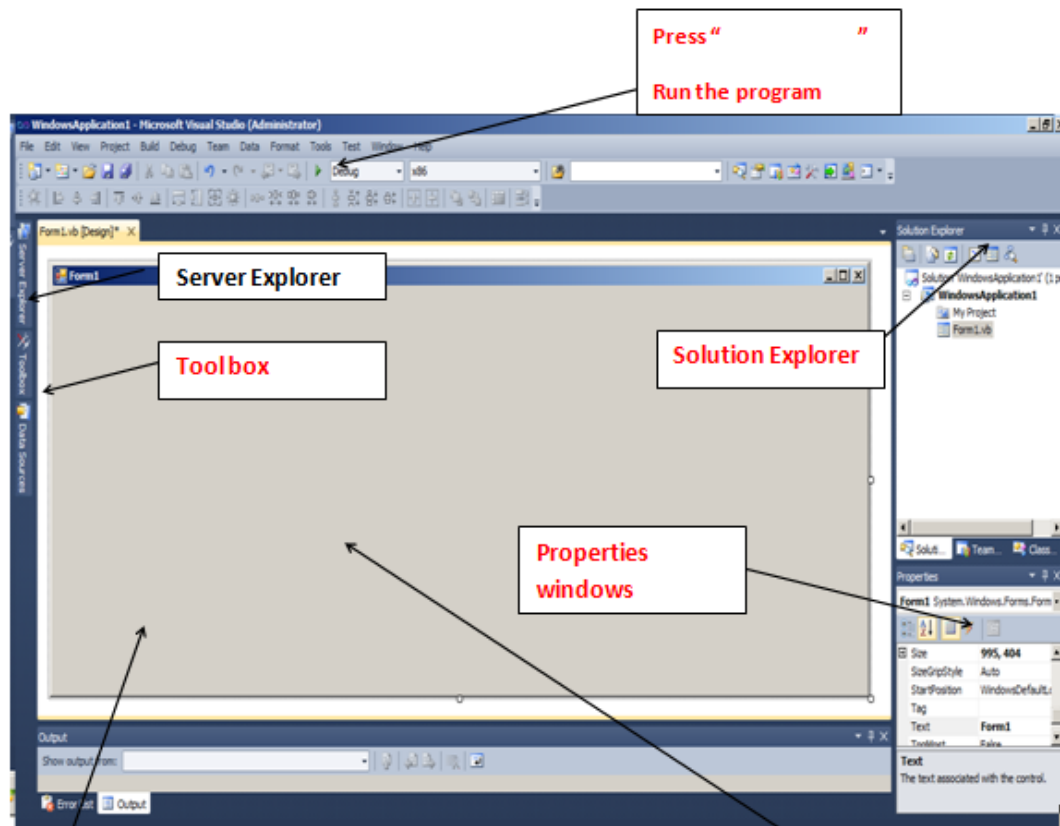
Click the Windows Forms Application icon in the central Templates area of the dialog box.

Visual Studio prepares the development environment for Visual Basic Windows application programming.

Visual basic → and choose → Windows forms Application

In the Name text box, type **first program**

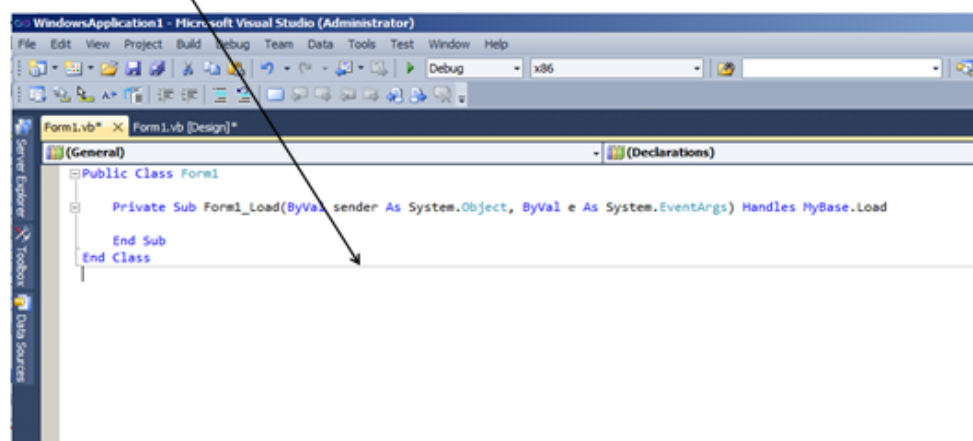
Click OK to create the new project in Visual Studio.



In the windows form design page and appeared in Left and side for Tool box option

If you want coding window, use the mouse on "Double click" and show on coding windows or "Right click → choose "view code"

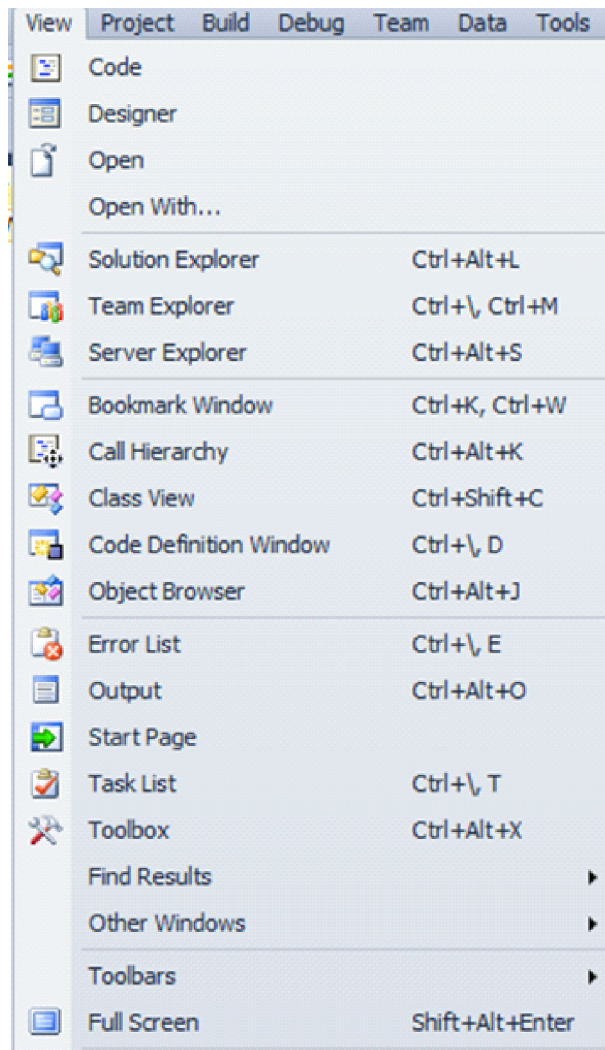
Design page



The Important of short keys are followed using in the keyboard.

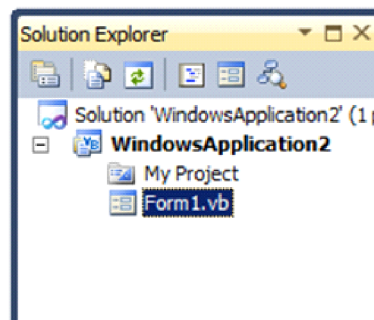
Tool box	→	CTRL+ALT+X
Solution Explorer	→	CTRL+ALT+L
Properties windows	→	F4 (Function key)
Run the program	→	F5

If you want mouse options, you go to the menu bar and select the view option.



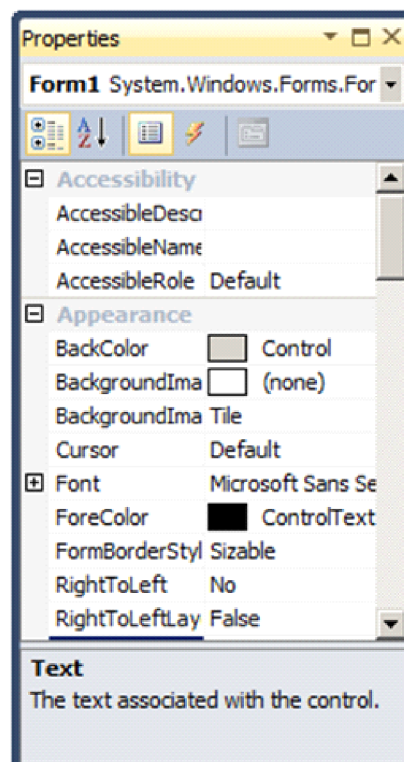
Solution Explorer

Solutions and projects contain items that represent the references, data connections, folders, and files that you need to create your application. A solution container can contain multiple projects and a project container typically contains multiple items.



Properties window

Each control we have on our user interface has lots of Properties we can set. This is done in the Properties window:

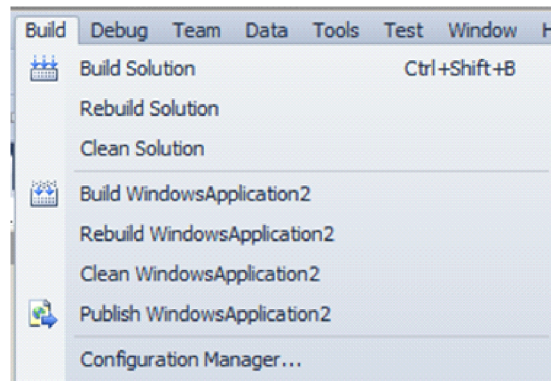


Build and Debug Tools

In Visual Studio we have lots of Build and Debugging Tools.

Buildmenu:

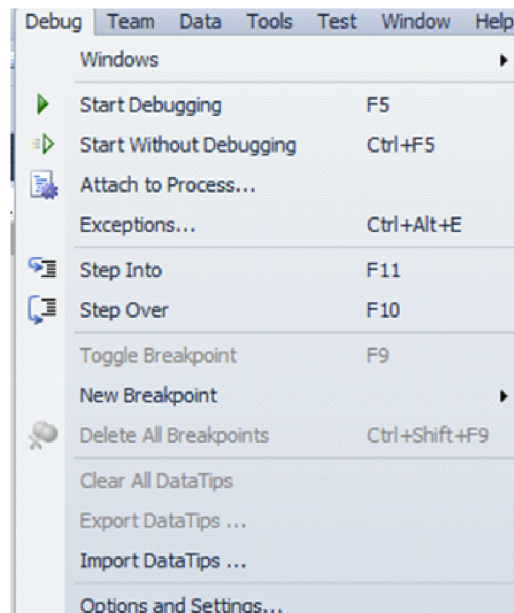
Below we see the **Build** menu:



The most used tool is “**BuildSolution**” (Shortcut Key: F6).

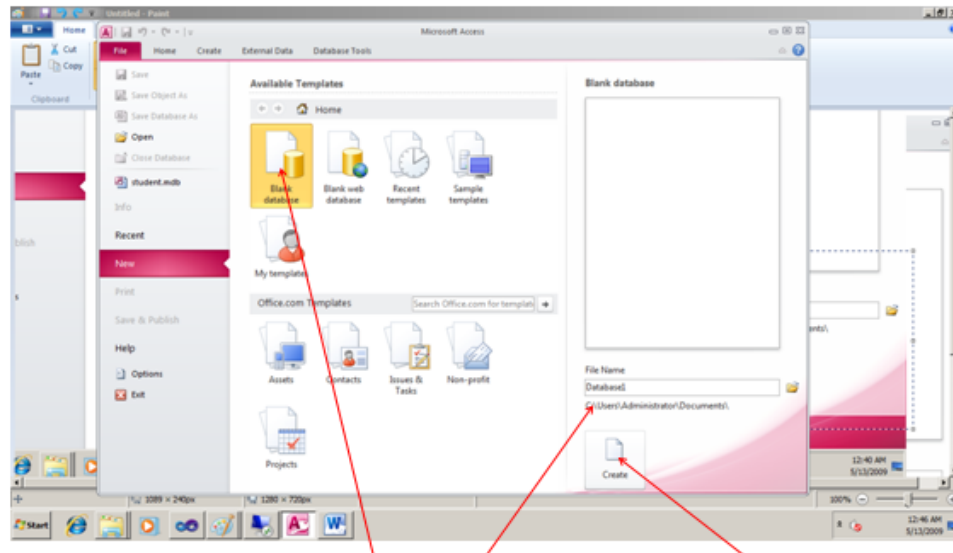
Debug menu:

Below we see the **Debug** menu:

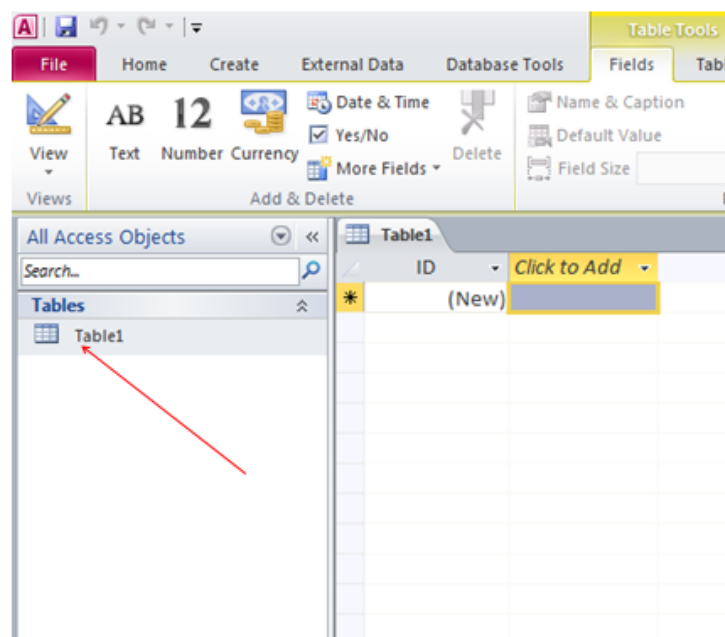


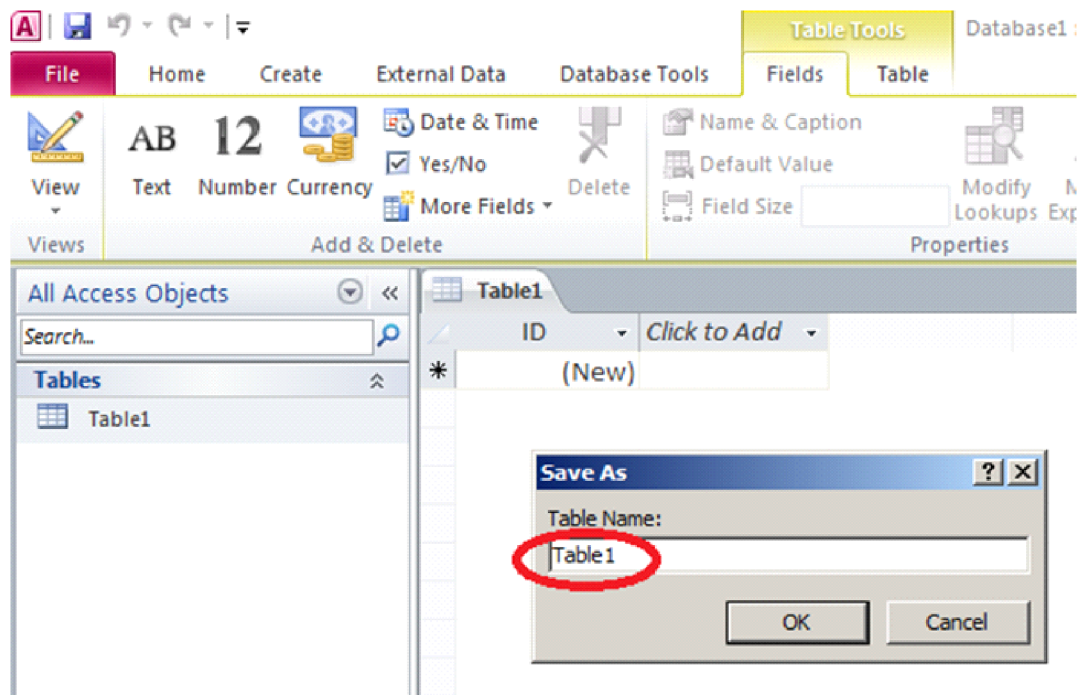
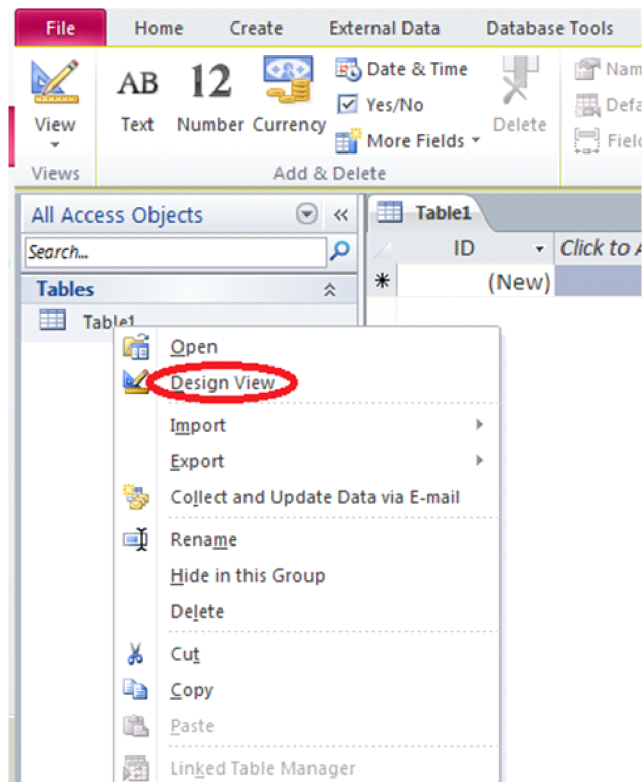
The most used tool is “**Start Debugging**” (Shortcut Key: F5).

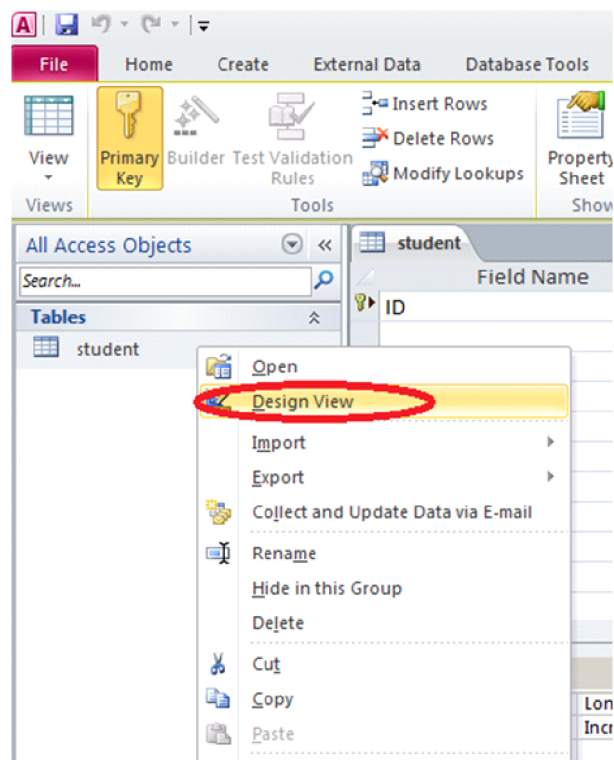
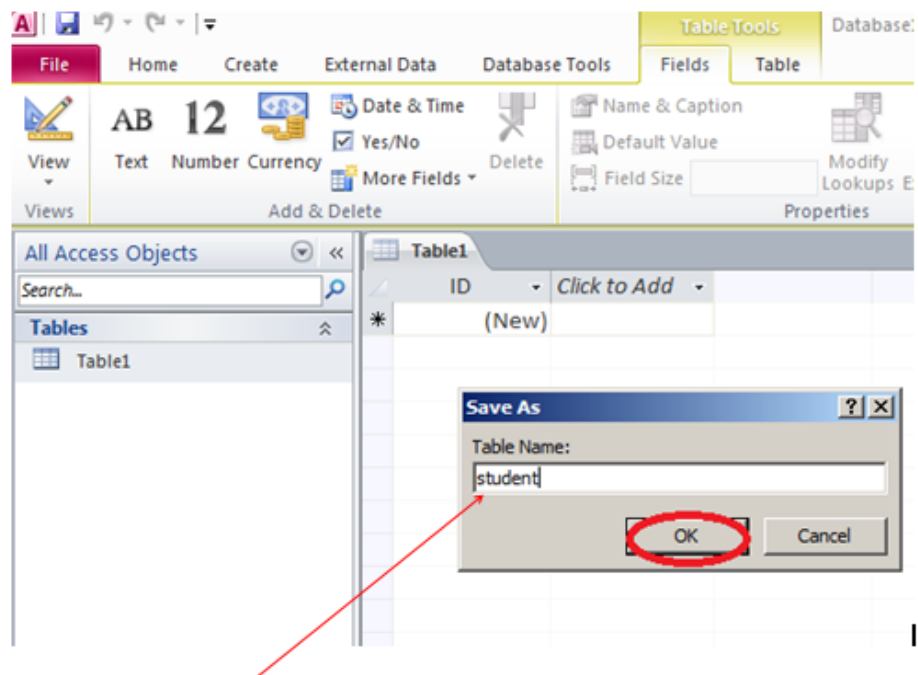
Back end Creation in Microsoft Access 2010

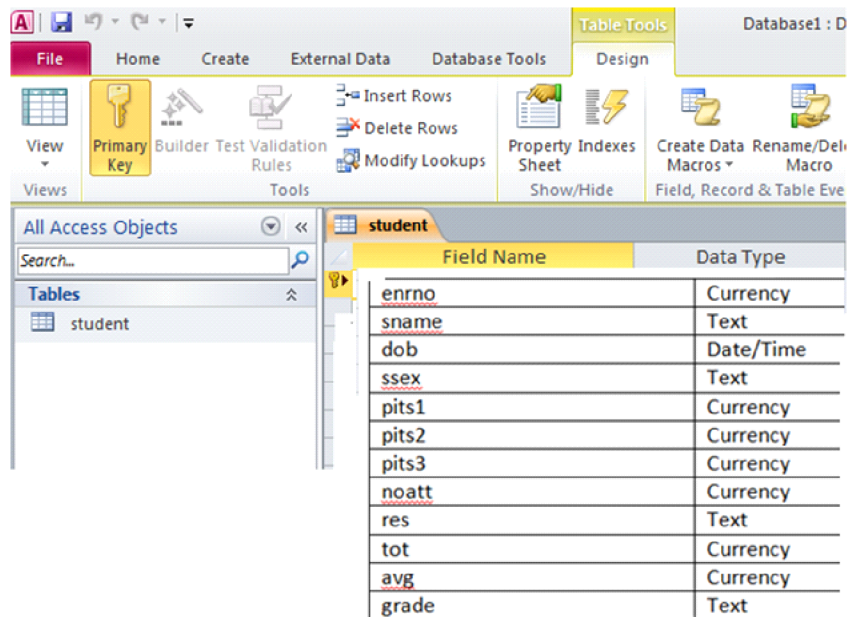
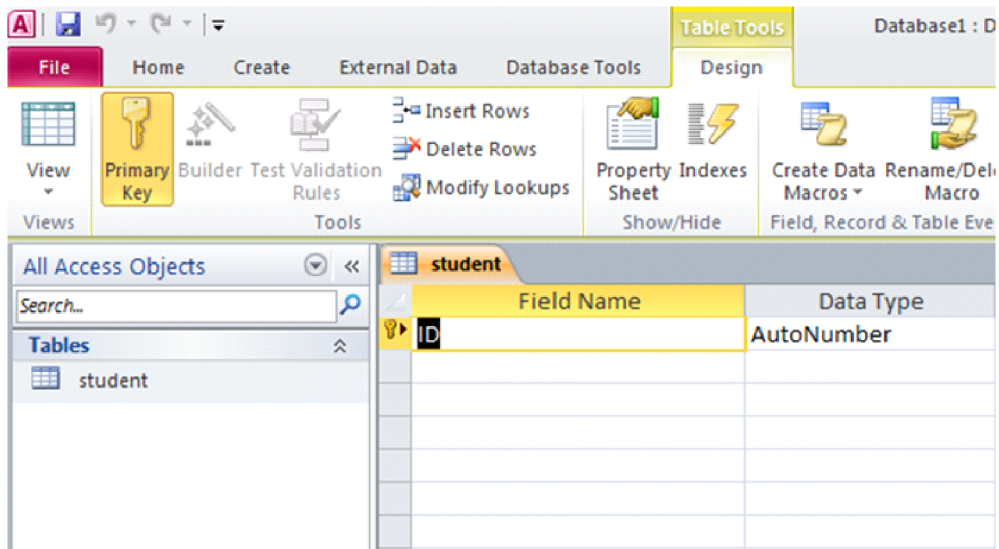


Your folder → DB creation of Location







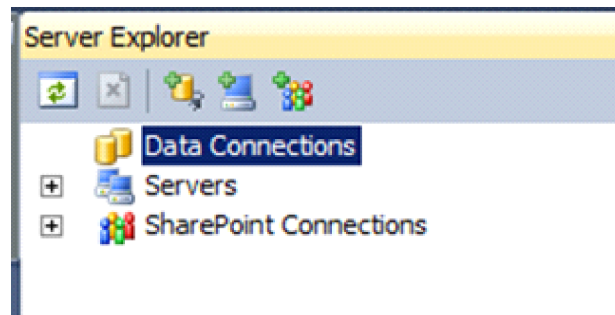


Front end and back end connection procedure

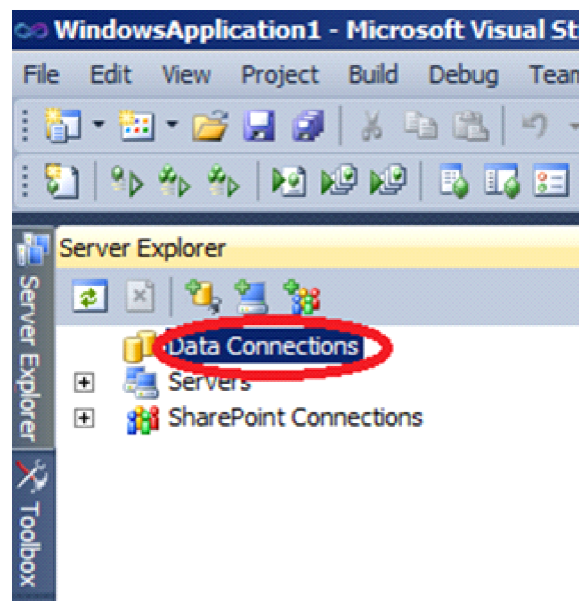
The following steps to be followed for connecting the backend to front end forms.

Select the server explorer option from the view menu of the standard tool bar. From where choose data connection.

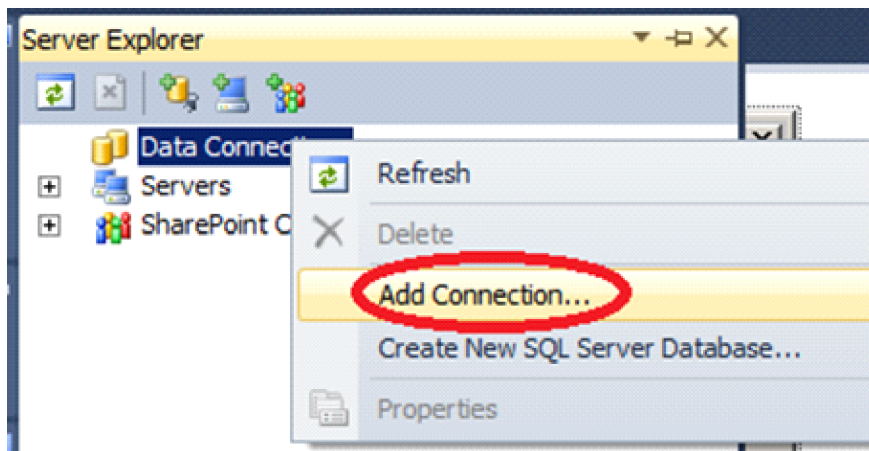
Step1: → go to Server explorer



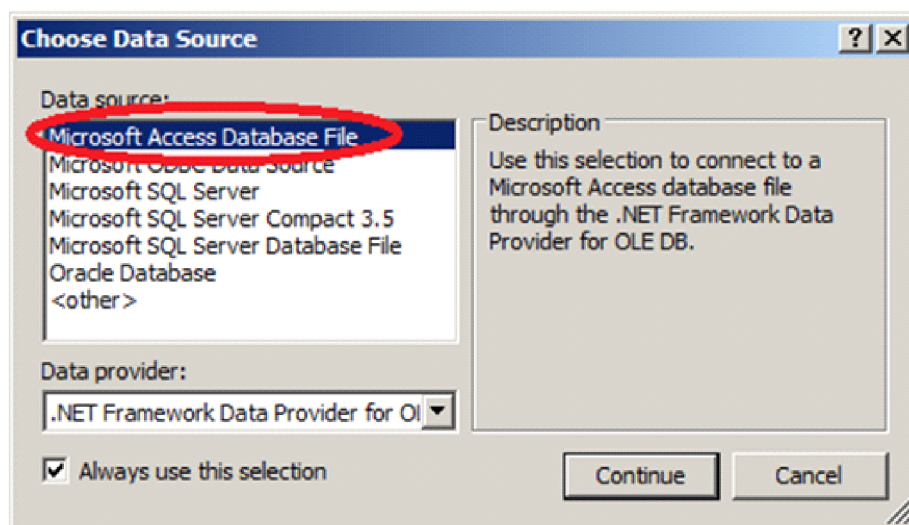
Choose → Data connections



Right click and choose → Add Connection



Next appear dialog box as follows àthen choose Data Source and select the→**Microsoft Access Database File**



Then "Continue" and choose "Browse" and select the DataBase

Add Connection [?] [X]

Enter information to connect to the selected data source or click "Change" to choose a different data source and/or provider.

Data source:
Microsoft Access Database File (OLE DB) [Change...]

Database file name:
[Browse...]

Log on to the database

User name: Admin

Password: []

☐ Save my password

[Advanced...]

[Test Connection] [OK] [Cancel]

Add Connection [?] [X]

Enter information to connect to the selected data source or click "Change" to choose a different data source and/or provider.

Data source:
Microsoft Access Database File

Database file name:
C:\Users\Administrator\Desktop

Log on to the database

User name: Admin

Password: []

☐ Save my password

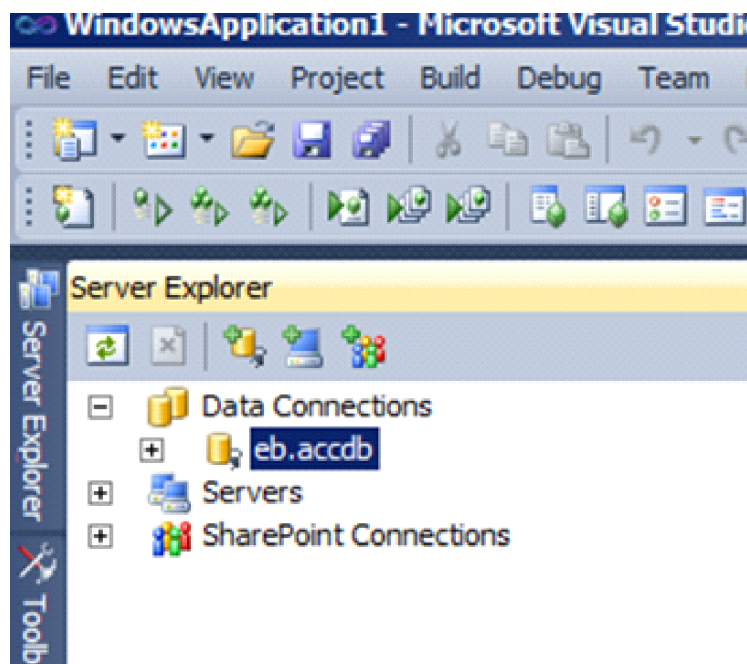
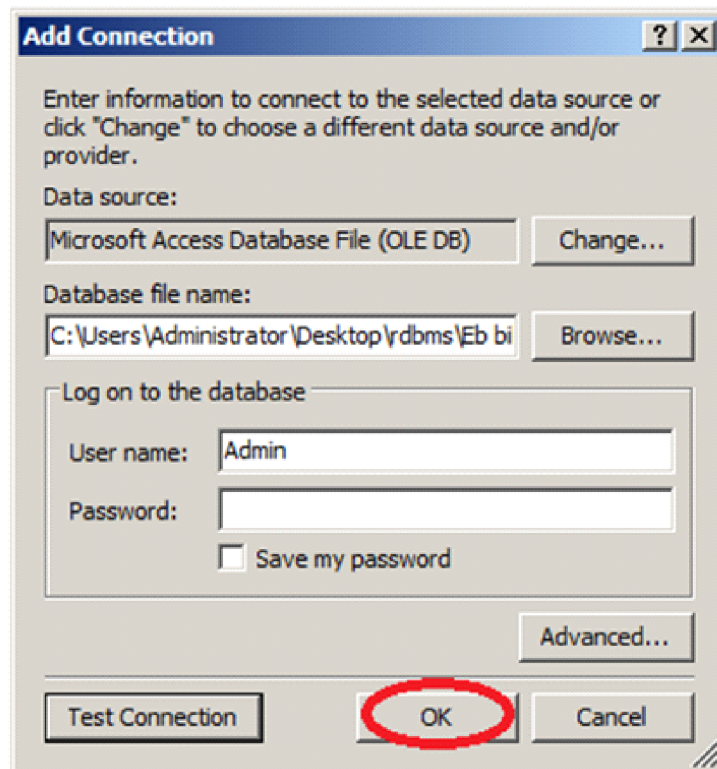
[Advanced...]

[Test Connection] [OK] [Cancel]

Microsoft Visual Studio [X]

Test connection succeeded.

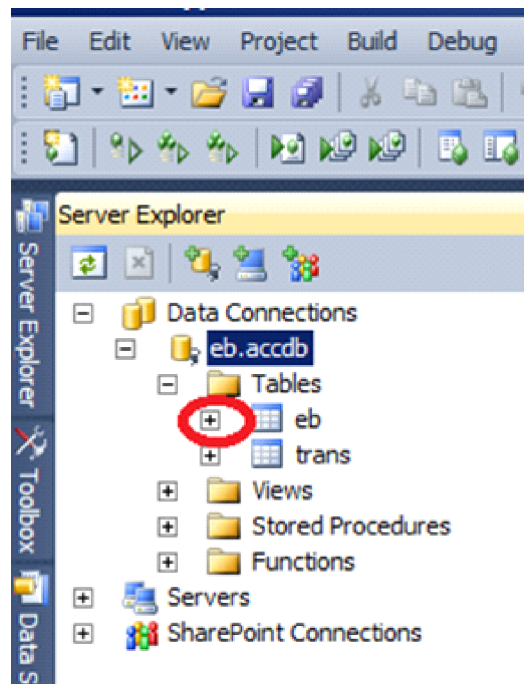
[OK]

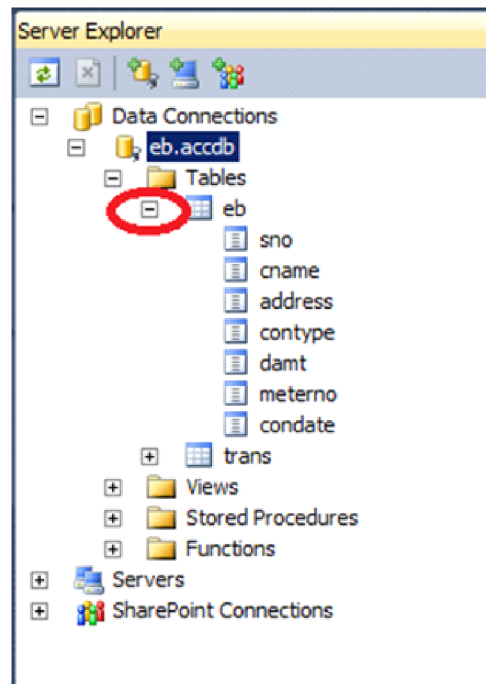




Right Click → and go to properties windows → and choose connection Strings → copy the path

Provider=Microsoft.ACE.OLEDB.12.0;Data Source=C:\Users\Administrator\rdbms\Eb bill\eb.accdb

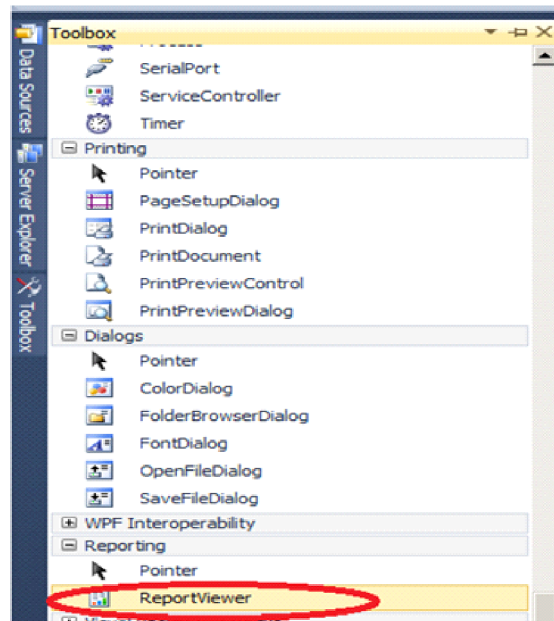




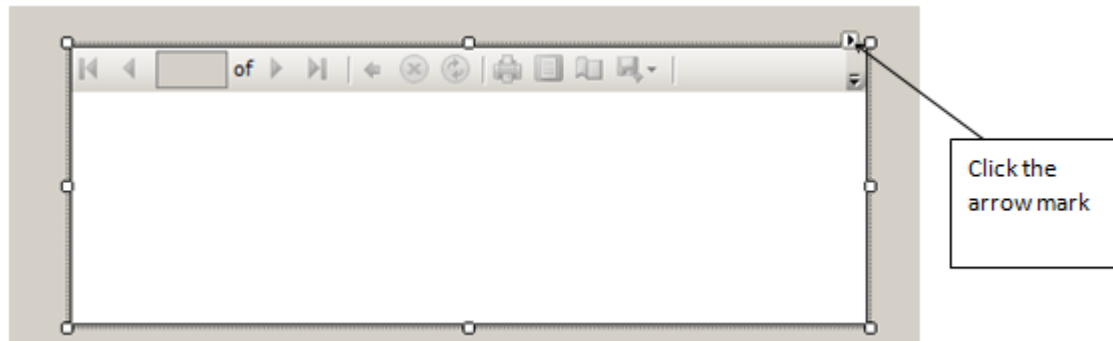
Creating a Report

Step 1:

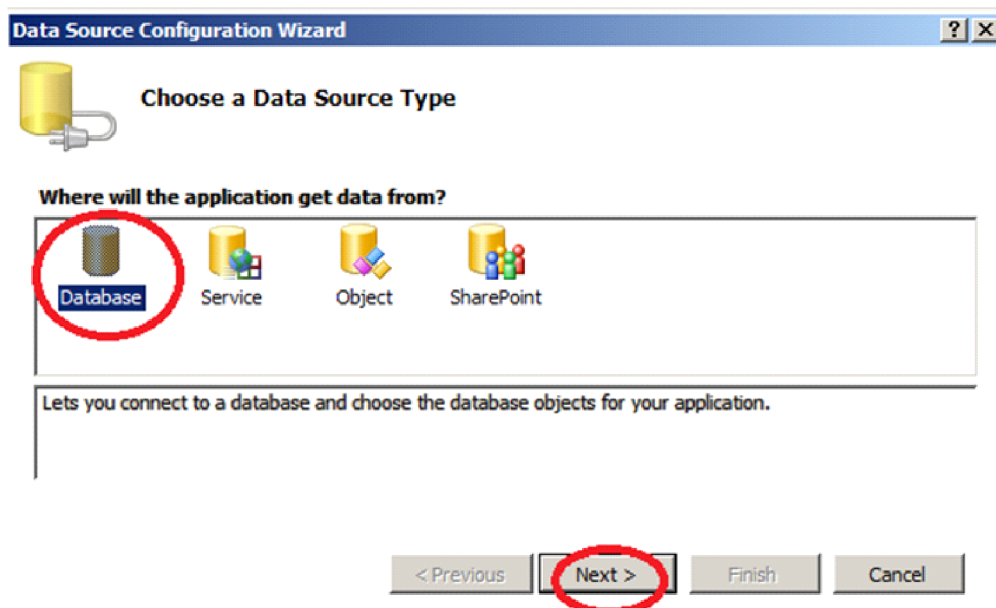
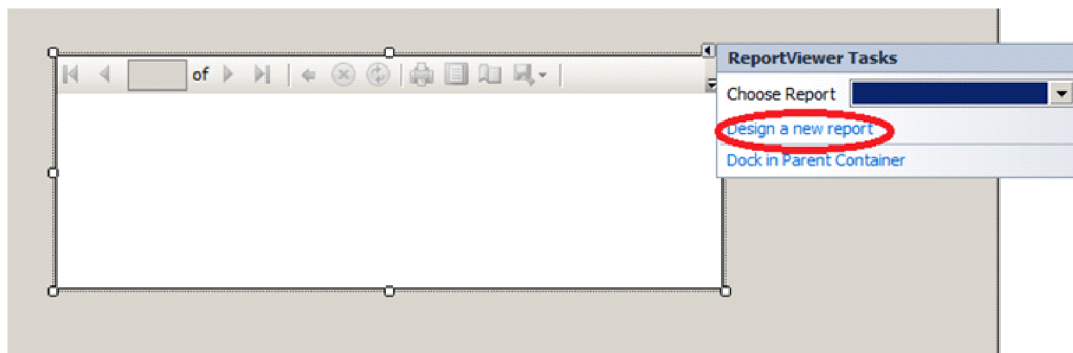
Goto toolbox and choose →reporting→ reportviewer



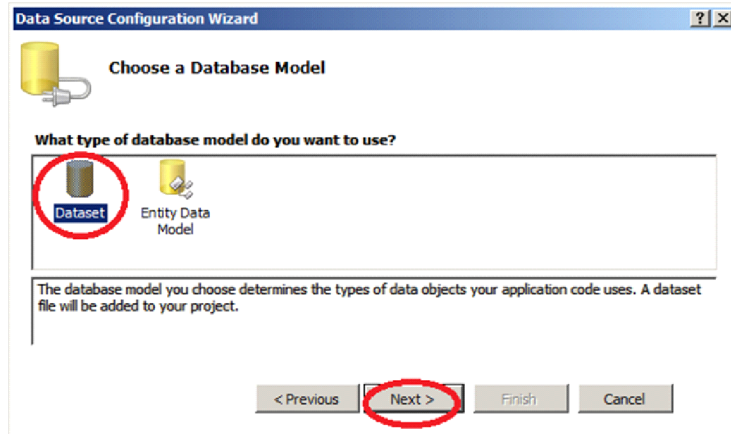
Create a New WindowsForm → Drag ReportViewer



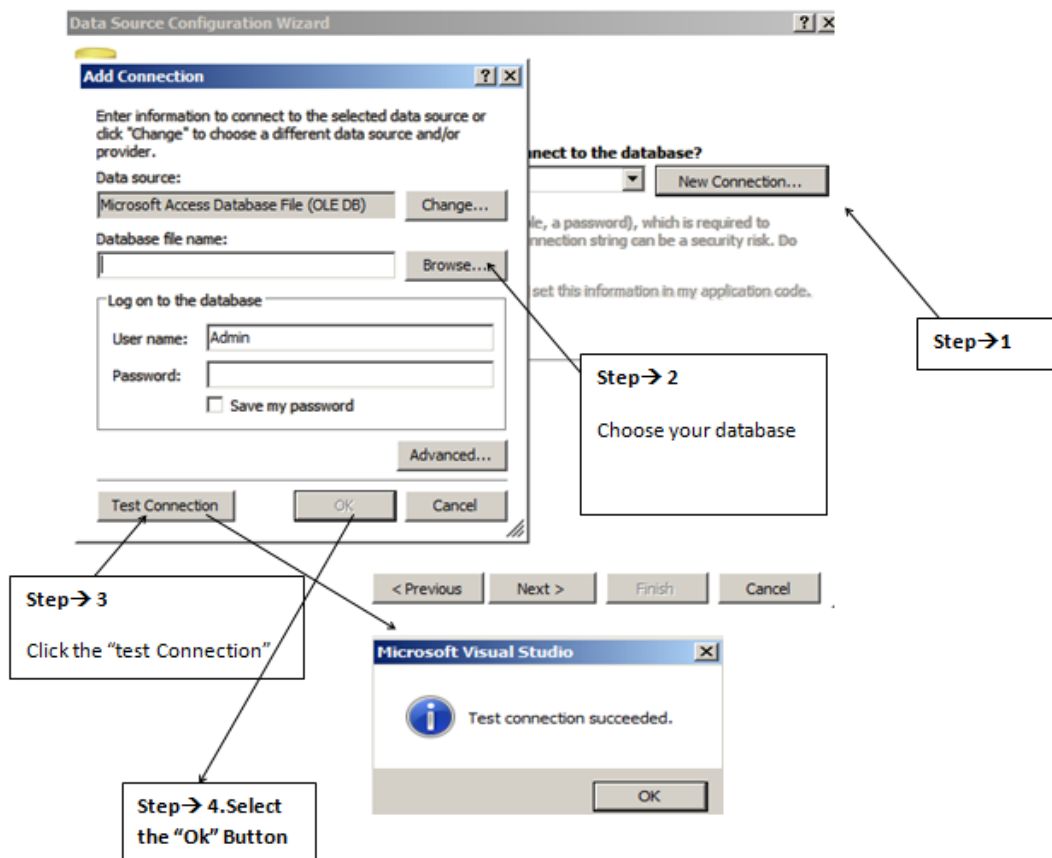
Next



Next



Next step



Next



Choose Your Data Connection

Which data connection should your application use to connect to the database?

eb.acddb New Connection...

This connection string appears to contain sensitive data (for example, a password), which is required to connect to the database. However, storing sensitive data in the connection string can be a security risk. Do you want to include this sensitive data in the connection string?

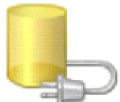
☐ No, exclude sensitive data from the connection string. I will set this information in my application code.

☐ Yes, include sensitive data in the connection string.

+ Connection string

< Previous **Next >** Finish Cancel

Next



Save the Connection String to the Application Configuration File

Storing connection strings in your application configuration file eases maintenance and deployment. To save the connection string in the application configuration file, enter a name in the box and then click Next.

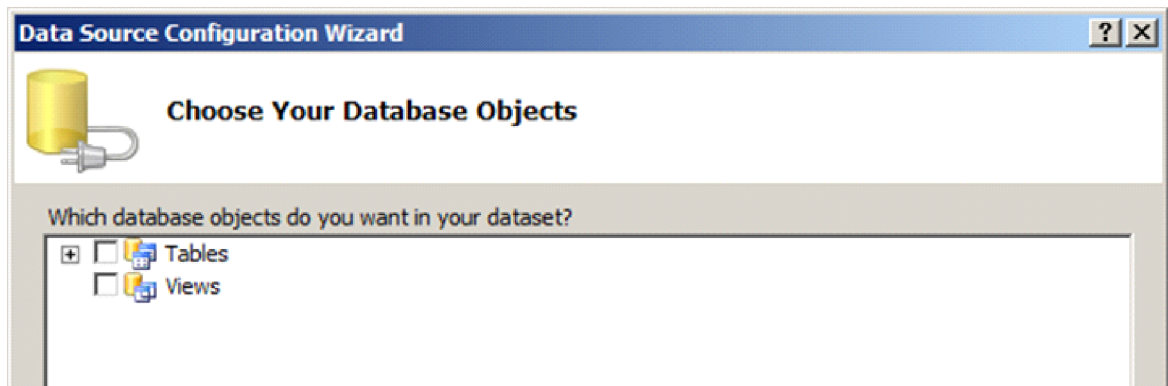
Do you want to save the connection string to the application configuration file?

☒ Yes, save the connection as:

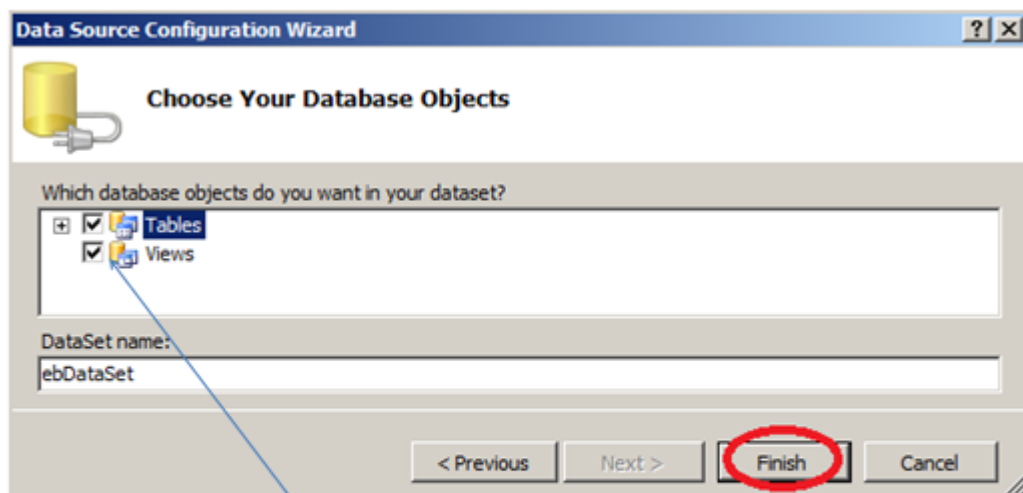
ebConnectionString

< Previous **Next >** Finish Cancel

Next



Next



Select the "tick mark" and
click the finish button

Report Wizard

Dataset Properties

Choose the Dataset

Name:

Data source:

Available datasets:

Fields:

Field Name	Type Name
consumerno	System.Decimal
cname	System.String
address	System.String
contype	System.String
damt	System.Decimal

Next

Report Wizard

Arrange fields

Arrange fields to group data in rows, columns, or both, and choose values to display. Data expands across the page in column groups and down the page in row groups. Use functions such as Sum, Avg, and Count on the fields in the Values box.

Available fields

- consumerno
- cname
- address
- contype
- damt

☐ Column groups

☐ Row groups

Σ Values

- Sum(consumerno)
- cname
- address
- contype

Select the required fields one by one drag on to → "Values" section

Next the layout will be displayed as follows

Report Wizard

Choose the layout

If you choose to show subtotals and grand totals, you can place them above or below the group. Stepped reports show hierarchical structure with indented groups in the same column.

Options:

- ☐ Show subtotals and grand totals
 - ☐ Blocked, subtotal below
 - ☐ Blocked, subtotal above
 - ☐ Stepped, subtotal above
- ☒ Expand/collapse groups

Preview

consumerno cname	address	contype
[Sum(consumer [cname])]	[address]	[contype]

< Back Next > Cancel

Report Wizard

Choose a style

Styles feature different fonts and color schemes, but do not affect the basic layout. You can customize the style after you finish the wizard.

Styles:

- Corporate**
- Forest
- Generic
- Mahogany
- Ocean
- Slate

Preview

consumerno cname	address	contype
[Sum(consumer [cname])]	[address]	[contype]

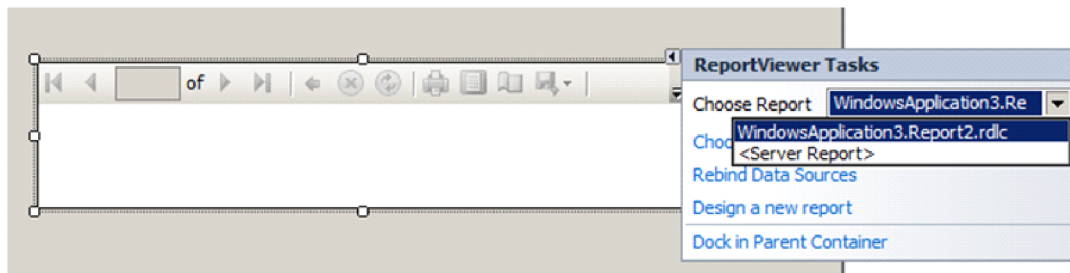
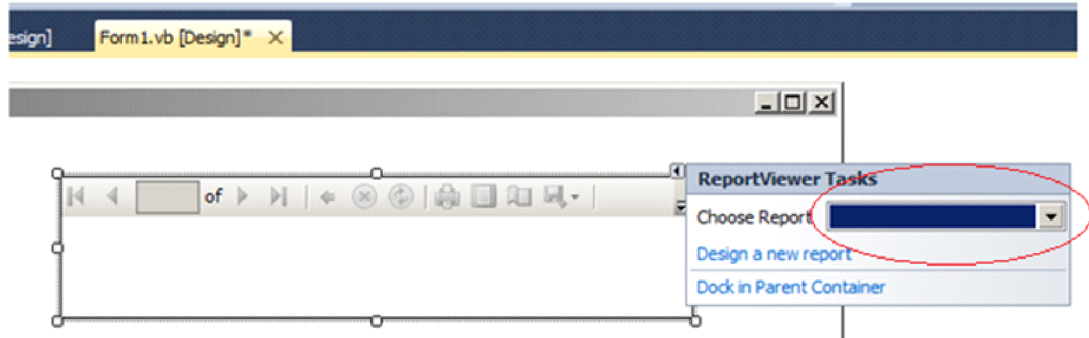
< Back **Finish >>** Cancel



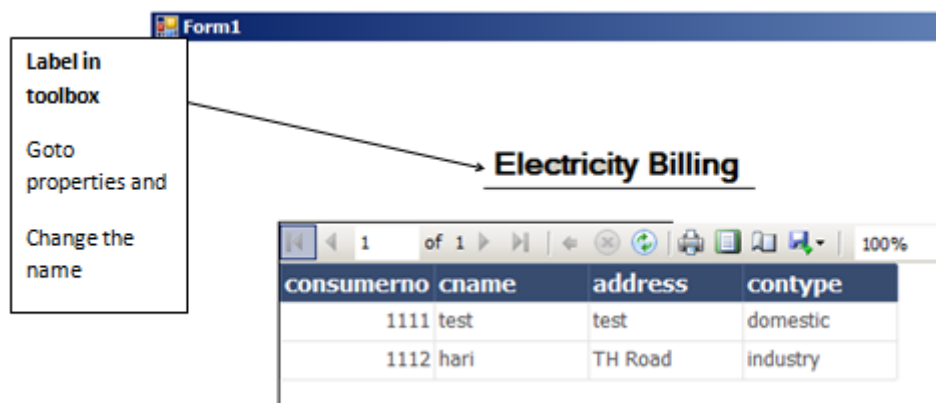
Now created

"Report1.rdlc" on the solution Explorer → next
goto Form1.vb[design]

next



Run the report form and get the report form



1. Aim : To create student database and produce Mark listfor the Students

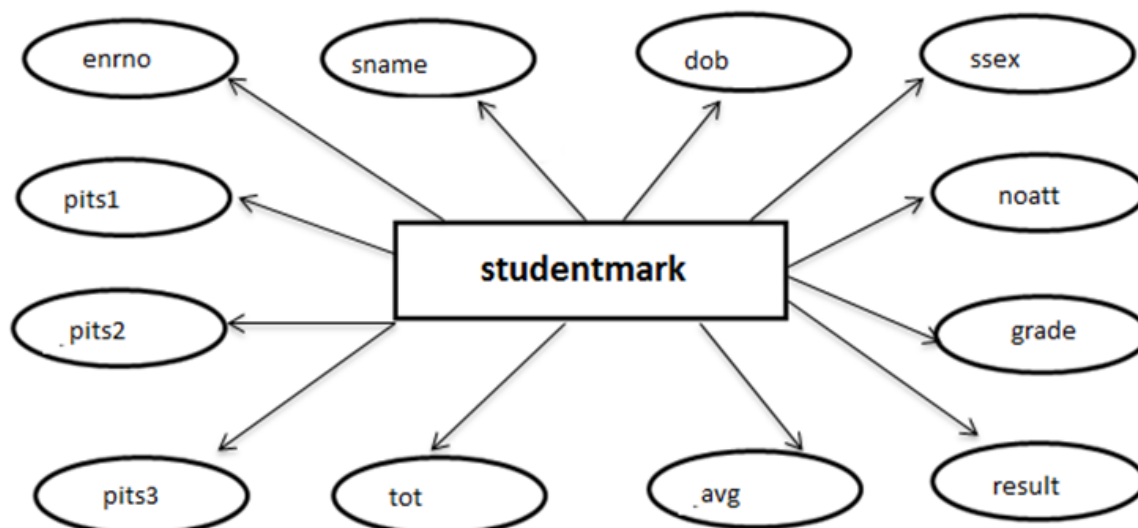
Procedure

- 1) Creating a new project in windows applications.
- 2) Front End Application using - VB .Net 2010
- 3) Back End Application using - Microsoft Access 2010
- 4) In FILE menu select "NEW PROJECT".
- 5) Give the name for Project "studentmarksheet" and click "OK". Window with Form1.vb gets generated.
- 6) Select textboxes, labels, menu strips, combo box and buttons fromToolbox.
- 7) Set the properties of controls as below.
- 8) Build the Project and Execute by Start Debugging or by pressing F5.

Table structure

Table structure : studentmark

Field Name	Type
enrno	Currency
sname	Text
dob	Date/Time
ssex	Text
pits1	Currency
pits2	Currency
pits3	Currency
noatt	Currency
res	Text
tot	Currency
avg	Currency
grade	Text

ER Diagram:**RESULT:**

Thus the program is compiled and executed successfully and the reports are obtained.

Student mark sheet processing

1. Designing the Form:
2. Step 1: Open Visual Studio 2010 → Windows Application
3. Step 2 : Create a menu use the Menu tooltip

Table structure :student

Field Name	Type	Description
enrno	Currency	Enrollment Number
sname	Text	Name of the Student
dob	Date/Time	Date of Birth
ssex	Text	Gender
pits1	Currency	Mark
pits2	Currency	Mark
pits3	Currency	Mark
noatt	Currency	No of attendance
res	Text	Result
tot	Currency	Total
avg	Currency	Average
grade	Text	Grade

Form design

Form design

University Of Chennai
Mark Entry Form

Enrollment No :

Student Name :

No Of Session Attended

Subject Code

Marks Obtained

OS

JAVA

C++

RDBMS

D.O.B

Gender

Total

Average

Grade

Calculation

Setting Properties

Controls name	Properties	values
Label1	text	Enrollment No
Label2	text	Student Name
Label3	text	No Of Session Attended
Label4	text	D.O.B
Label5	text	Gender
Label6	text	Subject Code
Label7	text	Marks Obtained
Label8	text	OS
Label9	text	JAVA
Label10	text	C++
Label11	text	RDBMS
Label12	text	Total

Label13	text	Average
Label14	text	Grade
Textbox1	name	tenrno
Textbox2	name	tsname
Textbox3	name	tdob
Textbox4	name	tpits1
Textbox5	name	tpits2
Textbox6	name	tpits3
Textbox7	name	tpits4
Textbox8	name	tatt1
Textbox9	name	ttot
Textbox10	name	tavg
Textbox11	name	tres
Combo box1	name	tssex
Combo box2	name	combroll
MenuStrip1		

Form1.vb Coding

Imports System.Data.OleDb

Public Class Form1

Dim con As New OleDbConnection

Dim cmd As New OleDbCommand

Dim dr As OleDbDataReader

Private Sub Form1_Load (ByVal sender As System.Object, ByVal e As System.EventArgs) Handles MyBase.Load

Btupdate.Visible = False

con =

NewOleDbConnection("Provider=Microsoft.Jet.OLEDB.4.0;

DataSource=C:\Users\Desktop\rdbms\

WindowsApplication2\WindowsApplication2\student.mdb")

```

con.Open()
MsgBox("DB Connected")
combroll.Visible = False
cmd.Connection = con
cmd.CommandText = "select max(enrno) from stu "
dr = cmd.ExecuteReader()
While dr.Read
    tenrno.Text = Val(dr(0).ToString()) + 1
End While
dr.Close()

```

End Sub

Private Sub Calcul_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles Calcul.Click

```

ttot.Text = Convert.ToInt32(tpits1.Text) + Convert.ToInt32(tpits2.Text) +
    Convert.ToInt32(tpits3.Text) + Convert.ToInt32(tpits4.Text)
tavg.Text = Val(ttot.Text) / 4
If Val(tpits1.Text) < 50 Or Val(tpits2.Text) < 50 Or Val(tpits3.Text) < 50 Or Val(tpits4.Text)
    < 50 Then
    tres.Text = "fail"
Elseif Val(tavg.Text) > 49 And Val(tavg.Text) < 60 Then
    tres.Text = "Second"
Elseif Val(tavg.Text) > 60 And Val(tavg.Text) < 70 Then
    tres.Text = "First"
Elseif Val(tavg.Text) > 70 Then
    tres.Text = "Distinction"
End If
End Sub

```

Private Sub ExitToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles ExitToolStripMenuItem.Click

End

End Sub

Private Sub AddRecordToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles AddRecordToolStripMenuItem.Click

Dim result As String

cmd.Connection = con

Btupdate.Visible = False

If (Val(tpits1.Text) < 50) Or Val(tpits2.Text) < 50 Or Val(tpits3.Text) < 50 Or
Val(tpits4.Text) < 50 Then

result = "Fail"

Elseif (Val(tpits1.Text) > 49 Or Val(tpits2.Text) > 49 Or Val(tpits3.Text) > 49 Or
Val(tpits4.Text) > 49) Then

result = "Pass"

Elseif tpits1.Text = "AAA" Or tpits2.Text = "AAA" Or tpits3.Text = "AAA" Or tpits4.Text =
"AAA" Then

result = "AAAA"

End If

cmd.CommandText = "insert into stuvalues(@enrno,@sname,@dob,@ssex,
@pits1,@pits2, @pits3,@pits4,@noatt1,@res, @tot,@avg,@grade)"

cmd.Parameters.AddWithValue("@enrno", tenrno.Text)

cmd.Parameters.AddWithValue("@sname", tsname.Text)

cmd.Parameters.AddWithValue("@dob", Convert.ToDateTime(tdob.Text))

cmd.Parameters.AddWithValue("@ssex", tssex.Text)

cmd.Parameters.AddWithValue("@pits1", Convert.ToInt32(tpits1.Text))

cmd.Parameters.AddWithValue("@pits2", Convert.ToInt32(tpits2.Text))

cmd.Parameters.AddWithValue("@pits3", Convert.ToInt32(tpits3.Text))

cmd.Parameters.AddWithValue("@pits4", Convert.ToInt32(tpits4.Text))

cmd.Parameters.AddWithValue("@noatt1", Convert.ToInt32(tatt1.Text))

```

cmd.Parameters.AddWithValue("@res", result.ToString())
cmd.Parameters.AddWithValue("@tot", Convert.ToInt32(ttot.Text))
cmd.Parameters.AddWithValue("@avg", tavg.Text)
cmd.Parameters.AddWithValue("@grade", tres.Text)
cmd.ExecuteNonQuery()
MsgBox("Record Added Successfully")
cmd.Clone()
con.Close()

```

End Sub

Private Sub ClearRecordToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles ClearRecordToolStripMenuItem.Click

```

tenrno.Text = ""
tsname.Text = ""
tatt1.Text = ""
tdob.Text = ""
tpits1.Text = ""
tpits2.Text = ""
tpits3.Text = ""
tpits4.Text = ""
ttot.Text = ""
tavg.Text = ""
tres.Text = ""

```

End Sub

Private Sub ReportToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles ReportToolStripMenuItem.Click

```

report.Show()
report.ReportViewer1.Show()

```

End Sub

**Private Sub tatt1_LostFocus(ByVal sender As Object, ByVal e As System.EventArgs)
Handles tatt1.LostFocus**

```

    Dim a As Double
    a = (Val(tatt1.Text) / 32 * 100)
    If a < 50.0# Then
        MsgBox("Attendance Lack")
        tpits1.Enabled = False
        tpits2.Enabled = False
        tpits3.Enabled = False
        tpits4.Enabled = False
        ttot.Enabled = False
        tavg.Enabled = False
        tres.Enabled = False
        Calcul.Enabled = False
        tpits1.Text = 0
        tpits2.Text = 0
        tpits3.Text = 0
        tpits4.Text = 0
        ttot.Text = 0
        tavg.Text = 0
        tres.Text = ""
    Else
        tpits1.Enabled = True
        tpits2.Enabled = True
        tpits3.Enabled = True
        tpits4.Enabled = True
        ttot.Enabled = True
        tavg.Enabled = True
        tres.Enabled = True
        Calcul.Enabled = True
    End If
End Sub

```

Private Sub DeleteRecordToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles DeleteRecordToolStripMenuItem.Click

```

    Dim n As String, f As Integer, dstr As String
    n = InputBox("Enter the Student Roll No to be Deleted")
    Btupdate.Visible = False
    cmd.Connection = con
    cmd.CommandText = "select * from stu"
    dr = cmd.ExecuteReader()
    While (dr.Read())
        If dr("enrno").ToString() = n Then
            cmd = New OleDbCommand("delete from stu where enrno =@enrno", con)
            cmd.Parameters.AddWithValue("@enrno", n)
            cmd.ExecuteNonQuery()
            MsgBox("record deleted ")
            f = 1
        End If
    End While
    dr.Close()
    If f = 0 Then
        MsgBox("Record not found")
    End If
    dr.Close()
End Sub

```

Private Sub UpdateRecordToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles UpdateRecordToolStripMenuItem.Click

```

    Btupdate.Visible = True
    combroll.Visible = True
    tenrno.Visible = False
    cmd.Connection = con

```

```

cmd.CommandText = "select * from stu"
dr = cmd.ExecuteReader()
While dr.Read()
    combroll.Items.Add(dr("enrno").ToString)
End While
dr.Close()
End Sub

```

Private Sub combroll_SelectedValueChanged(ByVal sender As Object, ByVal e As System.EventArgs) Handles combroll.SelectedValueChanged

```

cmd.Connection = con
cmd.CommandText = "select * from stu where enrno=@enrno"
cmd.Parameters.AddWithValue("@enrno", combroll.SelectedItem)
dr = cmd.ExecuteReader()
While dr.Read()
    tsname.Text = dr(1).ToString
    tdob.Text = dr(2).ToString
    tssex.Text = dr(3).ToString
    tpits1.Text = dr(4).ToString
    tpits2.Text = dr(5).ToString
    tpits3.Text = dr(6).ToString
    tpits4.Text = dr(7).ToString
    tatt1.Text = dr(8).ToString
End While
dr.Close()
End Sub

```

Private Sub Btupdate_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles Btupdate.Click

```

    Dim s As String
    s = "update stu set Sname=?,dob=?,SSex=?,pits1=?,pits2=?,pits3=?,
        pits4=?,Noatt1=?,tot=?,grade=? where enrno=?"
    cmd = New OleDbCommand(s, con)
    cmd.Parameters.AddWithValue("@sname", tsname.Text)
    cmd.Parameters.AddWithValue("@dob", Convert.ToDateTime(tdob.Text))
    cmd.Parameters.AddWithValue("@ssex", tssex.Text)
    cmd.Parameters.AddWithValue("@pits1", Convert.ToInt32(tpits1.Text))
    cmd.Parameters.AddWithValue("@pits2", Convert.ToInt32(tpits2.Text))
    cmd.Parameters.AddWithValue("@pits3", Convert.ToInt32(tpits3.Text))
    cmd.Parameters.AddWithValue("@pits4", Convert.ToInt32(tpits4.Text))
    cmd.Parameters.AddWithValue("@noatt1", Convert.ToInt32(tatt1.Text))
    cmd.Parameters.AddWithValue("@tot", Convert.ToInt32(ttot.Text))
    cmd.Parameters.AddWithValue("@grade", tres.Text)
    cmd.Parameters.AddWithValue("@enrno",
        Convert.ToInt32(combroll.SelectedItem))
    cmd.ExecuteNonQuery()
    MsgBox("Records Updated Successfully")
    Btupdate.Visible = False
End Sub
End Class

```

Output:

File Report Exit

University Of Chennai

Mark Entry Form

Enrollment No : D.O.B

Student Name : Gender

No Of Session Attended

Subject Code	Marks Obtained	Total
OS	65	276
JAVA	65	69
C++	68	First
RDBMS	78	

Report

University Of Chennai

Enrollment No	Name	Attendance	Gender	Total	Average	Grade
2	Rani	34	Female	276	69	First
3	Selvi	18	Male	312	78	Distinction
4	Raja	23	Female	291	72.75	Distinction
5	Hari	34	Male	257	64.25	First

2. Aim : To create Electricity master and transaction tables and produce the EB bill for the customer.

Table structure

Table Name: eb

ELECTRICITY BILL PROCESSING

Field Name	Type	Description
consumerno	Currency	Serial Number
Cname	Text	Name of the Customer
Address	Text	Address of the customer
Contype	Text	Connection Type
Damt	Currency	Deposit Amount

Table Name: trans

Field Name	Type	Description
Cname	Text	Customer Number
consumerno	Currency	Meter Number
Contype	Text	Connection Type
Preread	Currency	Previous Reading
Curread	Currency	Current Reading
Units	Currency	No of Units
Amount	Currency	Amount
Mrdate	Date/Time	Meter Reading Date

Form 1

Form1.vb [Design]

Form1.vb [Design]

File Monthly Reading Report Exit Type Here

Add
Clear
Delete
Type Here

TNEB CUSTOMER APPLICATION
North Chennai EB Distributed Circle

Consumer Number Customer Name

Address Connection Type

Deposit Amount

Coding**Form1.vb****Imports ADODB****Public Class Form1**

```
Dim con As New ADODB.Connection
```

```
Dim rs As New ADODB.Recordset
```

```
Dim cmd As New ADODB.Command
```

```
Dim flag As Integer = 1
```

```
Dim str As String
```

**Private Sub Form1_Load(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles MyBase.Load**

```
str = ""
```

```
combcontype.Items.Add("Domestic")
```

```
combcontype.Items.Add("Industry")
```

```
con.Open("Provider=Microsoft.ACE.OLEDB.12.0;DataSource=C:\Users\idell\Desktop\rdbms  
Eb bill\Eb bill\eb.accdb")
```

```
MsgBox("Connected")
```

```
cmd.ActiveConnection = con
```

```
cmd.CommandText = "select max(consumerno) from eb"
```

```
rs = cmd.Execute()
```

```
tconsumerno.Text = (Val(rs.Fields(0).Value) + 1).ToString
```

```
End Sub
```

**Private Sub AdToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e
As System.EventArgs) Handles AdToolStripMenuItem.Click**

```
'rs = con.Execute("insert into eb values (@sno)")
```

```
cmd.ActiveConnection = con
```

```
cmd.CommandText = "insert into eb
```

```
values(@consumerno,@cname,@address,@contype,@damt)"
```

```
cmd.Parameters.Item(0).Value = Convert.ToDouble(tconsumerno.Text)
```

```

cmd.Parameters.Item(1).Value = tcustname.Text
cmd.Parameters.Item(2).Value = taddress.Text
cmd.Parameters.Item(3).Value = combcontype.SelectedItem.ToString
cmd.Parameters.Item(4).Value = Convert.ToDouble(tdepamt.Text)
cmd.Execute()
MsgBox("Inserted")

```

End Sub

**Private Sub ReportToolStripMenuItem_Click(ByVal sender As System.Object,
ByVal e As System.EventArgs) Handles ReportToolStripMenuItem.Click**

```

Me.Hide()
Report.Show()
Report.ReportViewer2.Show()
Report.ReportViewer1.Hide()
Report.Label1.Visible = False
Report.Label2.Text = " EB Account Holder Report"

```

End Sub

**Private Sub ClearToolStripMenuItem_Click(ByVal sender As System.Object,
ByVal e As System.EventArgs) Handles ClearToolStripMenuItem.Click**

```

tconsumerno.Text = ""
tcustname.Text = ""
taddress.Text = ""
tdepamt.Text = ""

```

End Sub

**Private Sub ExitToolStripMenuItem_Click(ByVal sender As System.Object, ByVal
e As System.EventArgs) Handles ExitToolStripMenuItem.Click**

End Sub

```
Private Sub MonthlyReadingToolStripMenuItem_Click(ByVal sender As  

System.Object, ByVale As System.EventArgs) Handles  

MonthlyReadingToolStripMenuItem.Click
```

```
Me.Hide()
```

```
Read.Show()
```

```
End Sub
```

```
End Class
```

```
Private Sub DeleteToolStripMenuItem_Click(ByVal sender As System.Object,  

ByVal e As System.EventArgs) Handles DeleteToolStripMenuItem.Click
```

```
cmd.ActiveConnection = con
```

```
Dim no As Integer = Val(InputBox("Enter the Consumer no"))
```

```
cmd.CommandText = "select * from eb "
```

```
rs = cmd.Execute()
```

```
Do Until rs.EOF
```

```
    If rs.Fields(0).Value = no Then
```

```
        cmd.CommandText = "delete * from eb where sno=@sno"
```

```
        cmd.Parameters.Item(0).Value = no
```

```
        cmd.Execute()
```

```
        MsgBox("Record Deleted")
```

```
        flag = 0
```

```
        Exit Do
```

```
    End If
```

```
    rs.MoveNext()
```

```
Loop
```

```
If flag = 1 Then
```

```
    MsgBox("Record Not Found")
```

```
End If
```

```
End Sub
```

Form2:

Read

Update Record Bill Recipet Home

ELECTRICITY BILL PREPARATION

Reading Date	Previous Reading
Consumer Number	Current Reading
Customer Name	No Of Units
Connection Type	Amount

Read.vb**Public Class Read**

```
Dim con As New ADODB.Connection
```

```
Dim rs As New ADODB.Recordset
```

```
Dim cmd As New ADODB.Command
```

```
Dim flag As Integer = 1
```

```
Dim str As String
```

```
Private Sub Read_Load(ByVal sender As System.Object, ByVal e As  
System.EventArgs) Handles MyBase.Load
```

```
con.Open("Provider=Microsoft.ACE.OLEDB.12.0;Data  
Source=C:\Users\dell\Desktop\rdbms\Eb bill\Eb bill\eb.accdb")
```

```
Msgbox"DB connected"
```

```
treaddate.Text = Date.Now
```

```
treaddate.Select()
```

```
End Sub
```

Private Sub UpdateRecordToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles UpdateRecordToolStripMenuItem.Click

```

    cmd.ActiveConnection = con
cmd.CommandText = "insert into trans values
    (@cname,@consumemo,@contype,@preread,@curread,@units,@amount,@mrdate)"
    cmd.Parameters.Item(0).Value = Val(tconsumer.Text)
    cmd.Parameters.Item(1).Value = tcustname.Text
    cmd.Parameters.Item(2).Value = tcontype.Text
    cmd.Parameters.Item(3).Value = Val(tpread.Text)
    cmd.Parameters.Item(4).Value = Val(tcurread.Text)
    cmd.Parameters.Item(5).Value = Val(tunits.Text)
    cmd.Parameters.Item(6).Value = Val(tamount.Text)
    cmd.Parameters.Item(7).Value = CDate(treaddate.Text)
    cmd.Execute()
    MsgBox(" Record Updated")
End Sub

```

Private Sub tcurread_LostFocus(ByVal sender As Object, ByVal e As System.EventArgs) Handles tcurread.LostFocus

```

tunits.Text = (Convert.ToDouble(tcurread.Text) -
    Convert.ToDouble(tpread.Text)).ToString

```

End Sub

Private Sub tunits_LostFocus(ByVal sender As Object, ByVal e As System.EventArgs) Handles tunits.LostFocus

```

If tcontype.Text = "domestic" Then
    If Val(tunits.Text) < 100 Then
        tamount.Text = 2 * Val(tunits.Text)
    ElseIf Val(tunits.Text) > 100 And Val(tunits.Text) < 200 Then

```

```

        tamount.Text = 100 + (Val(tunits.Text) - 100) * 3
    ElseIf Val(tunits.Text) > 200 Then
        tamount.Text = 100 + 300 + (Val(tunits.Text) - 200) * 5
    End If
End If

If tcontype.Text = "commerical" Then
    If Val(tunits.Text) < 100 Then
        tamount.Text = 4.5 * Val(tunits.Text)
    ElseIf Val(tunits.Text) > 100 And Val(tunits.Text) < 200 Then
        tamount.Text = 450 + (Val(tunits.Text) - 100) * 3.5
    ElseIf Val(tunits.Text) > 200 Then
        tamount.Text = 450 + 750 + (Val(tunits.Text) - 200) * 10
    End If
End If

End Sub

```

```

Private Sub BillRecipetToolStripMenuItem_Click(ByVal sender As
System.Object, ByVal e As System.EventArgs) Handles
BillRecipetToolStripMenuItem.Click

```

```

    Me.Hide()
    Report.Show()
    Report.ReportViewer1.Show()
    Report.ReportViewer2.Hide()
    Report.Label1.Visible = True
    Report.Label2.Text = " Bill Receipt"

End Sub

```

```
Private Sub HomeToolStripMenuItem_Click(ByVal sender As System.Object,  
    ByVal e As System.EventArgs) Handles HomeToolStripMenuItem.Click
```

```
    Me.Hide()
```

```
    Form1.Show()
```

```
End Sub
```

```
Private Sub tconsumer_LostFocus(ByVal sender As Object, ByVal e As  
    System.EventArgs) Handles tconsumer.LostFocus
```

```
    cmd.ActiveConnection = con
```

```
    cmd.CommandText = "select * from eb where consumerno=@consumerno"
```

```
    cmd.Parameters.Item(0).Value = tconsumer.Text
```

```
    rs = cmd.Execute()
```

```
    Do Until rs.EOF
```

```
        tcustname.Text = rs.Fields(1).Value
```

```
        tcontype.Text = rs.Fields(3).Value
```

```
        rs.MoveNext()
```

```
        flag = 0
```

```
    Loop
```

```
    If flag = 1 Then
```

```
        MsgBox("Enter Valid Consumer Number")
```

```
    End If
```

```
End Sub
```

```
End Class
```

Output

FileMonthly ReadingReportExit

TNEB CUSTOMER APPLICATION
North Chennai EB Distributed Circle

Consumer Number1111

Customer NameHari

AddressMain Road

Connection TypeDomestic

Deposit Amount3333

Eb bill

Inserted

OK

Read

Update RecordBill RecipetHome

ELECTRICITY BILL PREPARATION

Reading Date6/30/2018 12:57:13 PM

Previous Reading34

Consumer Number1111

Current Reading67

Customer NameHari

No Of Units33

Connection Typedomestic

Amount66

Report

Report

Home

Bill Receipt

Name	Consumerno	Con Type	Previous Read	Current Read	Units	Amount
hari	1111	domestic	34	67	33	66
selvi	1112	domestic	67	123	56	256
raja	1113	domestic	456	567	111	133
rani	1114	domestic	123	444	321	1005

Signature

Report

Home

EB Account Holder Report

Consumerno	Name	Address	Con Type	Deposit Amount
1111	hari	Main Road	domestic	3333
1112	selvi	TH Road	industry	100000
1113	raja	KK Nagar	domestic	2000
1114	rani	Ashok Nagar	domestic	4000

4. Aim : To create master and transaction tables for Gas connection and Booking respectively. Process and generate Report for the same.

Table Structure

GAS BOOKING

Field Name	Type	Description
custno	Currency	Customer Number
custname	Text	Name of the Customer
custaddr	Text	Address of the customer
phone	Currency	Phone Number
nocylin	Currency	No of cylinder
damt	Currency	Deposit Amount

Table Name: trans

Field Name	Type	Description
custno	Currency	Customer Number
custname	Text	Name of the Customer
address	Text	Address of the customer
cphone	Currency	Phone Number
bookingno	Currency	Cylinder Booking Number
bdate	Date/Time	Booking Date
ddate	Date/Time	Delivery Date
billamt	Currency	Bill Amount

Form1

Form1

Master File Transaction Exit

Domestic Gas Customer Card

Customer No

Customer Name

Phone Number

Address

No Of Cylinder

Deposit

Update

Imports ADODB

Public Class Form1

Dim con As New ADODB.Connection

Dim rs As New ADODB.Recordset

Dim cmd As New ADODB.Command

Dim flag As Integer = 1

Dim no As Double

Private Sub Form1_Load(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles MyBase.Load

con.Open("Provider=Microsoft.ACE.OLEDB.12.0;Data
Source=C:\Users\dell\Desktop\rdbms\Gasbooking\gas.accdb")

bupdate.Visible = False

MsgBox("Connected")

End Sub

Private Sub AddToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles AddToolStripMenuItem.Click

bupdate.Visible = False

cmd.ActiveConnection = con

cmd.CommandText = "insert into gas
values(@custno,@custname,@custaddr,@phone,@nocylin,@damt)"

cmd.Parameters.Item(0).Value = Val(tcustno.Text)

cmd.Parameters.Item(1).Value = tcustname.Text

cmd.Parameters.Item(2).Value = taddress.Text

cmd.Parameters.Item(3).Value = Val(tpho.Text)

cmd.Parameters.Item(4).Value = Val(tcylin.Text)

cmd.Parameters.Item(5).Value = Val(tdept.Text)

cmd.Execute()

MsgBox("Added")

End Sub

```
Private Sub DeleteToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles DeleteToolStripMenuItem.Click
```

```
    bupdate.Visible = False
```

```
    cmd.ActiveConnection = con
```

```
    no = Val(InputBox("Enter the Customer No "))
```

```
    disp()
```

```
    If flag = 0 Then
```

```
        cmd.CommandText = "delete * from gas where custno=@custno"
```

```
        cmd.Parameters.Item(0).Value = no
```

```
        rs = cmd.Execute()
```

```
        MsgBox(" Record Deleted")
```

```
    End If
```

```
End Sub
```

```
Private Sub ExitToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles ExitToolStripMenuItem.Click
```

```
    End
```

```
End Sub
```

```
Private Sub UpdateToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e
    As System.EventArgs) Handles UpdateToolStripMenuItem.Click
```

```
    no = Val(InputBox("Enter customer Customer No"))
```

```
    disp()
```

```
    bupdate.Visible = True
```

```
End Sub
```

```
Public Function disp()
```

```
    cmd.ActiveConnection = con
```

```
    cmd.CommandText = "select * from gas where custno=@custno"
```

```
    cmd.Parameters.Item(0).Value = no
```

```
    rs = cmd.Execute()
```

Do Until rs.EOF

tcustno.Text = Val(rs.Fields(0).Value)

tcustname.Text = rs.Fields(1).Value

taddress.Text = rs.Fields(2).Value

tpho.Text = Val(rs.Fields(3).Value)

tcylin.Text = Val(rs.Fields(4).Value)

tdept.Text = Val(rs.Fields(5).Value)

rs.MoveNext()

flag = 0

Loop

If flag = 1 Then

MsgBox(" Invalid Customer No")

bupdate.Visible = False

End If

End Function

Private Sub bupdate_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles bupdate.Click

cmd.CommandText = "update gas set

custno=?,custname=?,custaddr=?,nocylin=?,damt=? where phone=?"

cmd.Parameters.Item(0).Value = Val(tcustno.Text)

cmd.Parameters.Item(1).Value = tcustname.Text

cmd.Parameters.Item(2).Value = taddress.Text

cmd.Parameters.Item(4).Value = Val(tcylin.Text)

cmd.Parameters.Item(5).Value = Val(tdept.Text)

cmd.Parameters.Item(3).Value = Val(tpho.Text)

cmd.Execute()

MsgBox(" Record Updated")

bupdate.Visible = False

End Sub

```

Private Sub TranscationToolStripMenuItem_Click(ByVal sender As System.Object,
        ByVal e As System.EventArgs) Handles TranscationToolStripMenuItem.Click
    Me.Hide()
    trans.Show()
End Sub
End Class
Trans.vb (coding)

```

Booking Bill report Home			
Gas Booking And Delivery Bill			
Booking No		Customer No	
Phone Number		Customer Name	
Address		Booking Date	
Delivery Date		Bill Amount	

Coding

Public Class trans

Dim con As New ADODB.Connection

Dim rs As New ADODB.Recordset

Dim cmd As New ADODB.Command

Dim flag As Integer = 1

Dim no As Double

```

Private Sub trans_Load(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles MyBase.Load
con.Open("Provider=Microsoft.ACE.OLEDB.12.0;Data
    Source=C:\Users\dell\Desktop\rdbms\Gasbooking\gas.accdb",
    UserID:"Admin", Password:="")
cmd.ActiveConnection = con
    cmd.CommandText = "select max(bookingno) from trans"
rs = cmd.Execute()
tbookno.Text = rs.Fields(0).Value + 1
End Sub

Private Sub BookingToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e
    As System.EventArgs) Handles BookingToolStripMenuItem.Click
cmd.ActiveConnection = con
cmd.CommandText = "insert into trans
    values(@custno,@custname,@custaddr,@phone,@booiingno,@bdate,ddate,billamt)"
cmd.Parameters.Item(0).Value = Val(tcustno.Text)
cmd.Parameters.Item(1).Value = tcustname.Text
cmd.Parameters.Item(2).Value = taddress.Text
cmd.Parameters.Item(3).Value = Val(tpho.Text)
cmd.Parameters.Item(4).Value = Val(tbookno.Text)
cmd.Parameters.Item(5).Value = Convert.ToDateTime(tbdate.Text)
cmd.Parameters.Item(6).Value = Convert.ToDateTime(tddate.Text)
cmd.Parameters.Item(7).Value = Val(tbamt.Text)
cmd.Execute()
MsgBox("Cylinder Booked")
clr()
End Sub

```

```
Private Sub HomeToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e
    As System.EventArgs) Handles HomeToolStripMenuItem.Click
```

```
    Me.Hide()
```

```
    Form1.Show()
```

```
End Sub
```

```
Function clr()
```

```
    tcustname.Text = ""
```

```
    tcustno.Text = ""
```

```
    taddress.Text = ""
```

```
    tpho.Text = ""
```

```
    tbookno.Text = ""
```

```
    tbamt.Text = ""
```

```
    tbdate.Text = ""
```

```
    tddate.Text = ""
```

```
End Function
```

```
Private Sub BillReportToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e
    As System.EventArgs) Handles BillReportToolStripMenuItem.Click
```

```
    Me.Hide()
```

```
    Report.Show()
```

```
    Report.DataGridView1.Show()
```

```
End Sub
```

```
Private Sub tcustno_LostFocus(ByVal sender As Object, ByVal e As
    System.EventArgs) Handles tcustno.LostFocus
```

```
    cmd.ActiveConnection = con
```

```
    cmd.CommandText = "select custname,phone,custaddr from gas where
        custno=?"
```

```
    cmd.Parameters.Item(0).Value = tcustno.Text
```

```
    rs = cmd.Execute()
```

```

Do Until rs.EOF
    tcustname.Text = rs.Fields(0).Value
    tpho.Text = rs.Fields(1).Value
    taddress.Text = rs.Fields(2).Value
    rs.MoveNext()
    flag = 0
    tbdate.Text = Date.Now
Loop
If flag = 1 Then
    MsgBox("Enter valid Customer Number")
End If
End Sub

Private Sub tbdate_LostFocus(ByVal sender As Object, ByVal e As System.EventArgs)
    Handles tbdate.LostFocus
    tddate.Text = Convert.ToDateTime(tbdate.Text).AddDays(5).ToString
    tbamt.Text = 700
End Sub
End Class

```

Output

The screenshot shows a Windows application window titled "Form1" with a menu bar containing "Master", "File", "Transaction", and "Exit". The main form is titled "Domestic Gas Customer Card" and contains six input fields arranged in two columns:

- Customer No: 123
- Customer Name: Hari
- Phone Number: 9787643
- Address: North Street
- No Of Cylinder: 2
- Deposit: 1000

A small dialog box titled "Gasbooking" is open on the right side of the form, displaying the text "Added" and an "OK" button.

trans

Booking

Bill report

Home

Gas Booking And Delivery Bill

Booking No

8

Customer No

123

Phone Number

9787643

Customer Name

hari

Address

North Street

Booking Date

6/28/2018 11:40:43 AM

Delivery Date

7/3/2018 11:39:39 AM

Bill Amount

700

Report

Report

Home

Gas Booking Bill Receipt

TNNO 12345

6/30/2018

	Name	Address	Phone No	Bill Date	Delivery Date	Bill Amount
▶	Hari	North Street	9787643	2/2/2018	2/3/2018	700
	raja	North Street	99354534	3/6/2018	3/11/2018	700
	selvi	south street	87651234	4/6/2018	4/11/2018	700
*						

Cylinder Weight : 39.9Kg

Signature

4. Aim : To create master and transation table for BANKaccount TRANSACTION and generate report.

Table Name : **master**

Field Name	Type	Description
accno	Currency	Account Number
cname	Text	Name of the Customer
address	Text	Address of the customer
phno	Currency	Phone Number
email	Text	Customer Email ID
acctype	Text	Account Type
bal	Currency	Account Balance
opdate	Date/Time	Account Opening Date

Table Name : **trans**

Field Name	Type	Description
accno	Currency	Account Number
cname	Text	Name of the Customer
address	Text	Address of the customer
transtype	Text	Transaction Type
tamt	Currency	Transaction Amount
tdate	Date/Time	Transaction Date

Form design

Form1

File Transcation Report Exit Type Here

Add
Delete
Update
Clear
Type Here

BANK OF UNIVERSE

Culomer Name Address

Proof Account Type

Email Id Date

Account No Balance

Update

Coding

```
Imports ADODB

Public Class Form1

    Dim con As New ADODB.Connection
    Dim rs As New ADODB.Recordset
    Dim cmd As New ADODB.Command
    Dim flag As Integer = 1
    Dim no As Double

    Private Sub Form1_Load(ByVal sender As System.Object, ByVal e As
        System.EventArgs) Handles MyBase.Load
con.Open("Provider=Microsoft.ACE.OLEDB.12.0;Data Source=C:\Users\dell\Desktop\
        rdbms\bank\bank\bank.accdb")

        MsgBox(" DB Connected")
        comboacctype.Items.Add("Saving")
        comboacctype.Items.Add("Current")
        comboproof.Items.Add("Aadhar Card")
        comboproof.Items.Add("Voter ID")
        comboproof.Items.Add("PAN Card")
        comboproof.Items.Add("Driving Lincense")
        btupdate.Visible = False
        tdate.Text = Date.Now
        tcustname.Select()
    End Sub

    Private Sub AddToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As
        System.EventArgs) Handles AddToolStripMenuItem.Click
        btupdate.Visible = False
        taddress.Enabled = True
        tdate.Enabled = True
        taccno.Enabled = True
    End Sub
End Class
```

```

cmd.ActiveConnection = con
cmd.CommandText = "insert into master values(@accno,@cname,
        @address,@phno,@email,@acctype,@bal,@opdate)"
cmd.Parameters.Item(0).Value = taccno.Text
cmd.Parameters.Item(1).Value = tcustname.Text
cmd.Parameters.Item(2).Value = taddress.Text
cmd.Parameters.Item(3).Value = comboproof.SelectedItem
cmd.Parameters.Item(4).Value = temail.Text
cmd.Parameters.Item(5).Value = comboacctype.SelectedItem
cmd.Parameters.Item(6).Value = tbal.Text
cmd.Parameters.Item(7).Value = tdate.Text
cmd.Execute()
MsgBox(" Record Added ")
End Sub

```

```

Private Sub DeleteToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e
        As System.EventArgs) Handles DeleteToolStripMenuItem.Click
    btupdate.Visible = False
    taddress.Enabled = True
    tdate.Enabled = True
    taccno.Enabled = True
    disp()
    If flag = 0 Then
        cmd.ActiveConnection = con
        cmd.CommandText = "delete * from master where accno=@accno"
        cmd.Parameters.Item(0).Value = no
        rs = cmd.Execute()
        MsgBox("Record Deleted")
    Else
        MsgBox("Record Not Deleted")
    End If

```

```

End If
End Sub
Public Function disp()
    cmd.ActiveConnection = con
    no = Val(InputBox("Enter the Account No"))
    cmd.CommandText = "select * from master where accno=@accno"
    cmd.Parameters.Item(0).Value = no
    rs = cmd.Execute()
    Do Until rs.EOF
        taccno.Text = rs.Fields(0).Value
        tcustname.Text = rs.Fields(1).Value
        taddress.Text = rs.Fields(2).Value
        comboproof.SelectedItem = rs.Fields(3).Value
        temail.Text = rs.Fields(4).Value
        comboacctype.SelectedItem = rs.Fields(5).Value
        tbal.Text = rs.Fields(6).Value
        tdate.Text = rs.Fields(7).Value
        rs.MoveNext()
        flag = 0
    Loop
    If flag = 1 Then
        MsgBox(" Record Not Found")
        btupdate.Visible = False
    End If
End Function
Public Function clr()
    tcustname.Text = ""
    taccno.Text = ""
    taddress.Text = ""

```

```

temail.Text = ""
tbal.Text = ""
tdate.Text = ""
End Function

```

```

Private Sub EditToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles EditToolStripMenuItem.Click
    disp()
    btupdate.Visible = True
    taddress.Enabled = False
    tdate.Enabled = False
    taccno.Enabled = False
End Sub

```

```

Private Sub btupdate_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles btupdate.Click
    cmd.ActiveConnection = con
    cmd.CommandText = "update master set
        cname=?,proof=?,email=?,acctype=?,bal=? where accno=?"
    cmd.Parameters.Item(0).Value = tcustname.Text
    cmd.Parameters.Item(1).Value = comboproof.SelectedItem
    cmd.Parameters.Item(3).Value = comboacctype.SelectedItem
    cmd.Parameters.Item(2).Value = temail.Text
    cmd.Parameters.Item(4).Value = Val(tbal.Text)
    cmd.Parameters.Item(5).Value = Val(taccno.Text)
    cmd.Execute()
    MsgBox(" record Updated")
    btupdate.Visible = False
End Sub

```

```

Private Sub UpdateToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e
    As System.EventArgs) Handles UpdateToolStripMenuItem.Click
    btupdate.Visible = False
    clr()
End Sub

```

```

Private Sub TranscationToolStripMenuItem_Click(ByVal sender As System.Object,
    ByVal e As System.EventArgs) Handles TranscationToolStripMenuItem.Click
    Me.Hide()
    trans.Show()
End Sub

```

```

Private Sub ReportToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e
    As System.EventArgs) Handles ReportToolStripMenuItem.Click
End Sub

```

```

Private Sub AccountsReportToolStripMenuItem_Click(ByVal sender As
    System.Object, ByVal e As System.EventArgs) Handles
    AccountsReportToolStripMenuItem.Click
    Me.Hide()
    Report.Show()
    Report.Label2.Text = "Account Holder Report"
    Report.ReportViewer1.Show()
    Report.ReportViewer2.Hide()
End Sub

```

```

Private Sub TranscationToolStripMenuItem1_Click(ByVal sender As System.Object,
    ByVal e As System.EventArgs) Handles TranscationToolStripMenuItem1.Click
    Me.Hide()
    Report.Show()

```

```

Report.Label2.Text = "Transaction Report"
Report.ReportViewer2.Show()
Report.ReportViewer1.Hide()
End Sub

```

```

Private Sub ExitToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles ExitToolStripMenuItem.Click
    End
End Sub

```

```

Private Sub temail_LostFocus(ByVal sender As Object, ByVal e As
    System.EventArgs) Handles temail.LostFocus
End Sub

```

```

Private Sub tdate_LostFocus(ByVal sender As Object, ByVal e As System.EventArgs)
    Handles tdate.LostFocus
    cmd.ActiveConnection = con
    cmd.CommandText = "select max(accno) from master"
    rs = cmd.Execute()
    taccno.Text = Val(rs.Fields(0).Value) + 1
    MsgBox(" Ur Account Number Generated")
    tbal.Text = 500
End Sub
End Class

```

Form2 design

The screenshot shows a Windows form titled "trans" with a blue title bar. The main content area has a white background with the text "BANK OF UNIVERSE" centered at the top. Below the title, there are seven input fields arranged in two columns. The left column contains "Account No", "Address", and "Date". The right column contains "Customer Name", "Operation Mode" (a dropdown menu), "Balance", and "Email Id". At the bottom of the form, there are four buttons: "Display", "Update", "Return", and "Clear", each in its own rectangular box.

Coding

Public Class trans

Dim con As New ADODB.Connection

Dim rs As New ADODB.Recordset

Dim cmd As New ADODB.Command

Dim flag As Integer = 1

Dim no As Double

Private Sub trans_Load(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles MyBase.Load

combmode.Items.Add("Withdraw")

combmode.Items.Add("Deposit")

con.Open("Provider=Microsoft.ACE.OLEDB.12.0;Data
Source=C:\Users\dell\Desktop\rdbms\bank\bank\bank.accdb")

tdate.Text = Date.Now

End Sub

```
Private Sub btDisplay_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles btDisplay.Click
```

```
    no = Val(InputBox("Enter the Account No"))
```

```
    disp()
```

```
End Sub
```

```
Public Function disp()
```

```
    cmd.ActiveConnection = con
```

```
    cmd.CommandText = "select * from master where accno=@accno"
```

```
    cmd.Parameters.Item(0).Value = no
```

```
    rs = cmd.Execute()
```

```
    Do Until rs.EOF
```

```
        taccno.Text = rs.Fields(0).Value
```

```
        tcustname.Text = rs.Fields(1).Value
```

```
        taddress.Text = rs.Fields(2).Value
```

```
        temail.Text = rs.Fields(4).Value
```

```
        tbal.Text = rs.Fields(6).Value
```

```
        rs.MoveNext()
```

```
        flag = 0
```

```
    Loop
```

```
    If flag = 1 Then
```

```
        MsgBox(" record Not Found")
```

```
    End If
```

```
End Function
```

```
Private Sub Btupdate_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles Btupdate.Click
```

```
    cmd.ActiveConnection = con
```

```
    cmd.CommandText = "update master set bal=? where accno=?"
```

```
    cmd.Parameters.Item(0).Value = temail.Text
```

```

cmd.Parameters.Item(1).Value = taccno.Text
cmd.Execute()
cmd.CommandText = "insert into trans
    (accno,cname,address,transtype,tamt,tdate) values
    (@accno,@cname,@address,@transtype,@tamt,tdate)"
cmd.Parameters.Item(0).Value = taccno.Text
cmd.Parameters.Item(1).Value = tcustname.Text
cmd.Parameters.Item(2).Value = taddress.Text
cmd.Parameters.Item(3).Value = combmode.SelectedItem
cmd.Parameters.Item(4).Value = tbal.Text
cmd.Parameters.Item(5).Value = tdate.Text
cmd.Execute()
MsgBox(" Record Updated")
End Sub

Private Sub combmode_SelectedIndexChanged(ByVal sender As System.Object,
    ByVal e As System.EventArgs) Handles combmode.SelectedIndexChanged
    Label7.Text = " Current Balance "
    Label6.Text = "Amount"
    temail.Text = tbal.Text
    tbal.Text = 0
End Sub

Private Sub tbal_LostFocus(ByVal sender As Object, ByVal e As System.EventArgs)
    Handles tbal.LostFocus
    If combmode.SelectedItem = "Deposit" Then
        temail.Text = Val(temail.Text) + Val(tbal.Text)
    Else
        If Val(temail.Text) < Val(tbal.Text) Then
            MsgBox(" Amount Not Suffient")
        Else

```

```

        temail.Text = Val(temail.Text) - Val(tbal.Text)
    End If
End If
End Sub

Private Sub btclear_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles btclear.Click
    taccno.Text = ""
    tcustname.Text = ""
    taddress.Text = ""
    temail.Text = ""
    tdate.Text = ""
    tbal.Text = ""
End Sub

Private Sub btreturn_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles btreturn.Click
    Me.Hide()
    Form1.Show()
End Sub

Private Sub taccno_LostFocus(ByVal sender As Object, ByVal e As
    System.EventArgs) Handles taccno.LostFocus
    no = Val(taccno.Text)
    disp()
End Sub
End Class

```

Output

Form1

FileTranscationReportExit

BANK OF UNIVERSE

Customer Name

Hari

Address

KK Nagar

Proof

PAN Card

Account Type

Saving

Email Id

hari@gmail.com

Date

6/29/2018 12:22:34 F

Account No

114

Balance

bank

Ur Account Number Generated

OK

trans

BANK OF UNIVERSE

Account No

114

Cutomer Name

Hari

Address

KK Nagar

Operation Mode

Deposit

Date

29/2018 12:24:18 PM

Amount

1000

Current Balance

1500

Display

Update

Return

Clear

bank

Record Updated

OK

Report

Report

Home

BANK OF UNIVERSE

Account Holder Report

Opening Date	Account Number	Name	Address	Proof	Acctype	Balance
6/21/2018 11:56:08 AM	111	selvi	TH Road	Aadhar Card	Saving	1400
6/22/2018 12:28:09 PM	112	Raja	North Street	PAN Card	Saving	10500
6/22/2018 1:45:27 PM	113	siva	MH Road	Aadhar Card	Saving	1500
6/29/2018 12:22:34 PM	114	Hari	KK Nagar	PAN Card	Saving	1500

Report

Home

BANK OF UNIVERSE

Transaction Report

Transaction Date	Account Number	Address	Transtype	Amount
3/3/2018 12:00:00 AM	111	North Street	Withdraw	1000
6/22/2018 12:30:44 PM	112	North Street	Deposit	500
6/22/2018 1:46:13 PM	114	MH Road	Deposit	1000
6/28/2018 11:03:25 AM	113	North Street	Deposit	10000

5. Aim: to create master and transaction table for maintaining the inventory system and generate the corresponding report

Table Name : master

Field Name	Type	Description
Itemcode	Currency	Item code
Itemname	Text	Name of the item
quantity	Currency	Quantity
pprice	Currency	Purchase Price
sprice	Currency	Sales Price
pdate	Date/Time	Purchase Date

Table Name : trans

Field Name	Type	Description
invno	Currency	Inventory Number
invdate	Date/Time	Inventory Date
custname	Text	Customer Name
address	Text	Address of the customer
phoneno	Currency	Phoneno
itemcode	Currency	Item Number
itemname	Text	Name of the item
unitprice	Currency	Unit
qty	Currency	No of quantity
amount	Currency	Amount
tax	Currency	Tax
totamt	Currency	Total Amount

Form Design

The screenshot shows a Windows application window titled 'Form1'. It has a menu bar with 'Purchase', 'Sales', 'Report', and 'Exit'. The main area is titled 'MADRAS TO CHENNAI STORE'. Below the title, there are six text input fields arranged in three rows and two columns. The first row contains 'Item Code' and 'Item Name'. The second row contains 'Quantity' and 'Purchase Price'. The third row contains 'Sales Price' and 'Purchase Date'. At the bottom of the window, there is a 'MenuStrip1' component.

Coding

```
Imports ADODB

Public Class Form1

    Dim con As New ADODB.Connection
    Dim rs As New ADODB.Recordset
    Dim cmd As New ADODB.Command
    Dim flag As Integer = 1
    Dim no As String

    Private Sub Form1_Load(ByVal sender As System.Object, ByVal e As
        System.EventArgs) Handles MyBase.Load
        con.Open("Provider=Microsoft.ACE.OLEDB.12.0;Data
            Source=C:\Users\del\Desktop\rdbms\inventory\inventory\inventory.accdb")
        MsgBox(" Connected")
    End Sub

    Private Sub AddToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As
        System.EventArgs) Handles AddToolStripMenuItem.Click
        cmd.ActiveConnection = con
        cmd.CommandText = "insert into master
            values(@itemcode,@itemname,@quantity,@pprice,@sprice,@pdate)"
        cmd.Parameters.Item(0).Value = ticode.Text
        cmd.Parameters.Item(1).Value = tiname.Text
        cmd.Parameters.Item(2).Value = tquty.Text
        cmd.Parameters.Item(3).Value = tpprice.Text
        cmd.Parameters.Item(4).Value = tspprice.Text
        cmd.Parameters.Item(5).Value = tupdate.Text
        cmd.Execute()
        MsgBox("Records Added Successfully")
        clr()
    End Sub
```

```
Public Function clr()
```

```
    ticode.Text = ""
```

```
    tiname.Text = ""
```

```
    tquty.Text = ""
```

```
    tpprice.Text = ""
```

```
    tsprice.Text = ""
```

```
    tupdate.Text = ""
```

```
End Function
```

```
Private Sub ViewToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e  
    As System.EventArgs) Handles ViewToolStripMenuItem.Click
```

```
    disp()
```

```
End Sub
```

```
Public Function disp()
```

```
    cmd.ActiveConnection = con
```

```
    no = InputBox("Enter the Itemname")
```

```
    cmd.CommandText = "select * from master where itemname=@itemname"
```

```
    cmd.Parameters.Item(0).Value = no
```

```
    rs = cmd.Execute()
```

```
    Do Until rs.EOF
```

```
        ticode.Text = rs.Fields.Item(0).Value
```

```
        tiname.Text = rs.Fields.Item(1).Value
```

```
        tquty.Text = rs.Fields.Item(2).Value
```

```
        tpprice.Text = rs.Fields.Item(3).Value
```

```
        tsprice.Text = rs.Fields.Item(4).Value
```

```
        tupdate.Text = rs.Fields.Item(5).Value
```

```
        flag = 0
```

```
        rs.MoveNext()
```

```
    Loop
```

```
    If flag = 1 Then
```

```
        MsgBox(" Record Not Found")
```

```

End If
End Function

Private Sub DeleteToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e
    As System.EventArgs) Handles DeleteToolStripMenuItem.Click
    disp()
    cmd.ActiveConnection = con
    cmd.CommandText = "delete * from master where itemname=@itemname"
    cmd.Parameters.Item(0).Value = no
    cmd.Execute()
    MsgBox(" Record Deleted Successfully")
    clr()
End Sub

Private Sub SalesToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e
    As System.EventArgs) Handles SalesToolStripMenuItem.Click
    Me.Hide()
    Sales.Show()
End Sub

Private Sub ReportToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e
    As System.EventArgs) Handles ReportToolStripMenuItem.Click
    Me.Hide()
    report.Show()
End Sub

Private Sub ExitToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles ExitToolStripMenuItem.Click
    End
End Sub
End Class

```

6. Aim: To create product and order tables and generate report for purchase order placed

Table Name : product

Field Name	Type	Description
pid	Currency	Purchase ID
pname	Text	Purchase Name
price	Currency	Price
soh	Currency	Stock on hand
pdate	Date/Time	Purchase Date

Table Name : orderdetails

Field Name	Type	Description
bname	Text	Buyer Name
organi	Text	Organisation
add	Text	Address
ordno	Currency	Order number
doord	Date/Time	Date of Order
tpid	Currency	Product ID
tpdesc	Text	Product Description
ordqty	Currency	Order Quantity
Unit	Currency	Unit Price
amount	Currency	Total Amount

Form design

The screenshot shows a Windows application window titled "purchase". Inside the window, the title "MADRAS TO CHENNAI STORE" is centered. Below the title, there are three input fields: "Purchase Date", "Supplier Name", and "Supplier Phone Number". Below these, there are five input fields arranged in a row: "Item Name", "Item Code", "Qty Nos", "Unit Price", and "Sales Price". At the bottom of the form, there are three buttons: "Bill Update", "Order Update", and "Return". A "MenuStrip1" is visible at the very bottom of the window.

Coding

```
Public Class purchase
```

```
    Dim con As New ADODB.Connection
```

```
    Dim rs As New ADODB.Recordset
```

```
    Dim cmd As New ADODB.Command
```

```
    Dim flag As Integer = 1
```

```
    Dim no As String
```

```
    Private Sub Button2_Click(ByVal sender As System.Object, ByVal e As  
        System.EventArgs) Handles btorder.Click
```

```
    End Sub
```

```
    Private Sub purchase_Load(ByVal sender As System.Object, ByVal e As  
        System.EventArgs) Handles MyBase.Load
```

```
        con.Open("Provider=Microsoft.ACE.OLEDB.12.0;Data  
            Source=C:\Users\del\Desktop\rdbms\inventory\inventory\inventory.accdb ")
```

```
    End Sub
```

```
    Private Sub btbill_Click(ByVal sender As System.Object, ByVal e As  
        System.EventArgs) Handles btbill.Click
```

```
        cmd.ActiveConnection = con
```

```
        cmd.CommandText = "insert into master
```

```
            values(@itemcode,@itemname,@pprice,@sprice,@pdate)"
```

```
        cmd.Parameters.Item(0).Value = ticode.Text
```

```
        cmd.Parameters.Item(1).Value = tiname.Text
```

```
        ' cmd.Parameters.Item(2).Value = tpprice.Text
```

```
        cmd.Parameters.Item(3).Value = tsprice.Text
```

```
        ' cmd.Parameters.Item(4).Value = tpdate.Text
```

```
        ' cmd.Execute()
```

```
        MsgBox("Records Added Successfully")
```

```
'clr()
```

```
End Sub
```

```
End Class
```

Sales form design

The screenshot shows a Windows application window titled "Sales". Inside the window, there is a form titled "Cash Bill" for "MADRAS TO CHENNAI STORE". The form includes the following elements:

- GST No:** ABD334FO
- INV NO:** A text input field.
- Customer Information:**
 - Customer Name:** A text input field.
 - Phone Number:** A text input field.
 - Address:** A text input field.
 - Sales Date:** A text input field.
- Table:** A table with 5 columns: Item Code, Item Description, Unit Price, Qty Nos, and Sales Price. Each column has a corresponding text input field below it.
- Summary:**
 - GST:** A text input field.
 - Total:** A text input field.
- Buttons:** Three buttons labeled "Sale", "Clear", and "Return" are located at the bottom left of the form.
- MenuStrip:** A menu strip labeled "MenuStrip1" is located at the bottom of the window.

Sales coding

```
Public Class Sales
```

```
Dim con As New ADODB.Connection
```

```
Dim rs As New ADODB.Recordset
```

```
Dim cmd As New ADODB.Command
```

```
Dim flag As Integer = 1
```

```
Dim no As String
```

```
Private Sub Sales_Load(ByVal sender As System.Object, ByVal e As  
System.EventArgs) Handles MyBase.Load
```

```
con.Open("Provider=Microsoft.ACE.OLEDB.12.0;Data
```

```

        Source=C:\Users\del\\Desktop\rdbms\inventory\inventory\inventory.accdb")
EndSub

PrivateSub btSale_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles btSale.Click
    cmd.ActiveConnection = con
    If flag = 0 Then
        cmd.CommandText = "insert into trans
            values(@invno,@invdate,@custname,@address,@phoneno,@itemcode,@
            itemname,@unitprice,@qty,@amount,@tax,@totamt)"
        cmd.Parameters.Item(0).Value = tinvno.Text
        cmd.Parameters.Item(1).Value = tsdate.Text
        cmd.Parameters.Item(2).Value = tcname.Text
        cmd.Parameters.Item(3).Value = taddress.Text
        cmd.Parameters.Item(4).Value = tphno.Text
        cmd.Parameters.Item(5).Value = ticode.Text
        cmd.Parameters.Item(6).Value = tiname.Text
        cmd.Parameters.Item(7).Value = tunit.Text
        cmd.Parameters.Item(8).Value = tqty.Text
        cmd.Parameters.Item(9).Value = tsprice.Text
        cmd.Parameters.Item(10).Value = tgst.Text
        cmd.Parameters.Item(11).Value = ttot.Text
        cmd.Execute()
        MsgBox(" Sales Successfully")
        clr()
    EndIf
EndSub

PublicFunction clr()
    tinvno.Text = ""
    tsdate.Text = ""

```

```

        tcname.Text = ""
        taddress.Text = ""
        tphno.Text = ""
        ticode.Text = ""
        tiname.Text = ""
        tunit.Text = ""
        tqty.Text = ""
        tsprice.Text = ""
        tgst.Text = ""
        ttot.Text = ""
    EndFunction

PrivateSub btclear_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles btclear.Click
    clr()
EndSub

PrivateSub btreturn_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles btreturn.Click
    Me.Hide()
    Form1.Show()
EndSub

PrivateSub tqty_LostFocus(ByVal sender As Object, ByVal e As System.EventArgs)
    Handles tqty.LostFocus
    cmd.ActiveConnection = con
    cmd.CommandText = "select quantity from master where itemcode=@itemcode "
    cmd.Parameters.Item(0).Value = ticode.Text
    rs = cmd.Execute()
DoUntil rs.EOF

```

```

If Val(tqty.Text) < Val(rs.Fields.Item(0).Value) Then
    cmd.CommandText = "update master set quantity=? where itemcode=?"
    cmd.Parameters.Item(0).Value = Val(rs.Fields.Item(0).Value) - Val(tqty.Text)
    cmd.Parameters.Item(1).Value = ticode.Text
    cmd.Execute()
    tsprice.Text = Val(tunit.Text) * Val(tqty.Text)
    tgst.Text = (Val(tsprice.Text) * 5) / 100
    ttot.Text = Val(tsprice.Text) + Val(tgst.Text)
    flag = 0
Else
    MsgBox("Stock not available")
    MsgBox("Stock on Hand "& Val(rs.Fields.Item(0).Value))

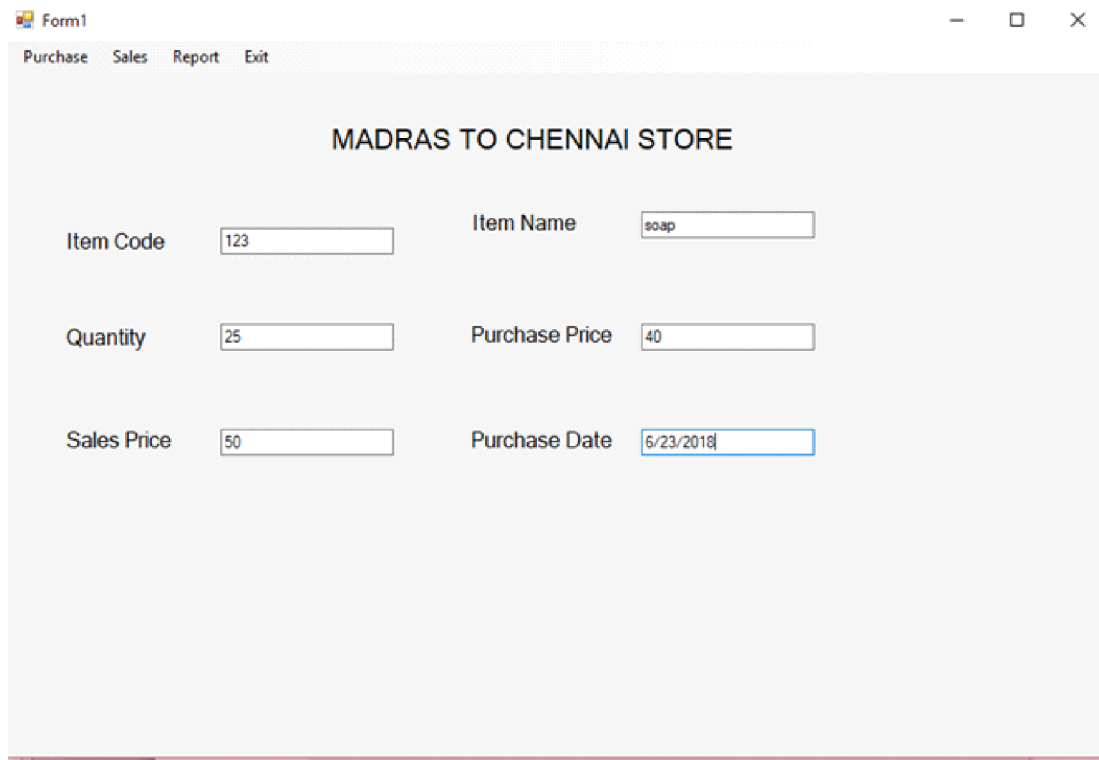
EndIf
rs.MoveNext()
Loop
EndSub

PrivateSub ticode_LostFocus(ByVal sender As Object, ByVal e As System.EventArgs)
    Handles ticode.LostFocus
    cmd.ActiveConnection = con
    cmd.CommandText = "select itemname,sprice from master where
        itemcode=@itemcode "
    cmd.Parameters.Item(0).Value = ticode.Text
    rs = cmd.Execute()
DoUntil rs.EOF
    tiname.Text = rs.Fields(0).Value
    tunit.Text = rs.Fields(1).Value
    flag = 0
    rs.MoveNext()

```

```
Loop
If flag = 1 Then
    MsgBox(" Invalid Item Code")
EndIf
EndSub
EndClass
```

Output



The screenshot displays a Windows application window titled "Form1". The window features a menu bar with the following options: "Purchase", "Sales", "Report", and "Exit". The main content area is titled "MADRAS TO CHENNAI STORE". Below the title, there are six input fields arranged in two columns and three rows:

Field Name	Value
Item Code	123
Item Name	soap
Quantity	25
Purchase Price	40
Sales Price	50
Purchase Date	6/23/2018

Sales

GST No ABD334FO

Cash Bill

INV NO 234

MADRAS TO CHENNAI STORE

Customer Name Raja Phone Number 98765444

Address Ashok Nagar Sales Date 6/12/2018

inventory

Sales Successfully

OK

Item Code	Item Description	Unit Price	Qty Nos	Sales Price
123	soap	50	2	100

GST : 5

Total 105

Sale Clear Return

Report

report

Home

MADRAS TO CHENNAI STORE

	ItemCode	ItemName	UnitPrice	Quantity	Amount
▶	112	soap	50	5	250
	111	pen	50	1	50
	113	oil	50	2	100
*					

7. Aim : To create table for employee personal and salary details and generate report for payroll

Table Name : emp

Field Name	Type	Description
empno	Currency	Employee Number
empname	Text	Employee Name
address	Text	Address of the employee
phone	Currency	Phone Number
desig	Text	Designation
department	Text	Name of the department
dob	Date/Time	Date of Birth
doj	Date/Time	Date of Joining
gender	Text	Gender
baccno	Currency	Bank Account No
basic	Currency	Basic Salary

Table Name : pay

Field Name	Type	Description
empno	Currency	Employee Number
empname	Text	Employee Name
desig	Text	Designation
phno	Currency	Phone Number
Basic	Currency	Basic Salary
Hra	Currency	HRA
Da	Currency	DA
Pf	Currency	PF
Gpay	Currency	Gross Pay
Netpay	Currency	Netpay
Cmonth	Currency	Month of Pay bill

Form Design

Form1

File Monthly Pay Report Exit Type Here

Add
Update
Find
Clear
Type Here

Payroll Processing

Employee No Employee Name

Date Of Birth Date Of Joining

Address Department Name

Phone Number Designation

Gender Basic

Bank Account No

Coding

Imports ADODB

Public Class combodesig

Dim con As New ADODB.Connection

Dim rs As New ADODB.Recordset

Dim cmd As New ADODB.Command

Dim flag As Integer = 1

Dim no As Double

Private Sub Form1_Load(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles MyBase.Load

con.Open("Provider=Microsoft.ACE.OLEDB.12.0;Data
Source=C:\Users\dell\Desktop\rdbms\payroll\payroll\payroll.accdb")

MsgBox(" Connected")

combosex.Items.Add("Male")

combosex.Items.Add("Female")

```

btupdate.Visible = False
cmd.ActiveConnection = con
cmd.CommandText = "select max(empno) from emp"
rs = cmd.Execute
tempno.Text = (rs.Fields(0).Value + 1)
End Sub

Private Sub AddToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles AddToolStripMenuItem.Click
    btupdate.Visible = False
    cmd.ActiveConnection = con
    cmd.CommandText = "insert into emp values (@empno, @empname, @address,
        @phone, @desig, @department, @dob, @doj, @gender, @baccno, @basic)"
    cmd.Parameters.Item(0).Value = tempno.Text
    cmd.Parameters.Item(1).Value = tempname.Text
    cmd.Parameters.Item(2).Value = taddress.Text
    cmd.Parameters.Item(3).Value = tpho.Text
    cmd.Parameters.Item(4).Value = tdesig.Text
    cmd.Parameters.Item(5).Value = tdept.Text
    cmd.Parameters.Item(6).Value = tdob.Text
    cmd.Parameters.Item(7).Value = tdoj.Text
    cmd.Parameters.Item(8).Value = combosex.SelectedItem
    cmd.Parameters.Item(9).Value = tbank.Text
    cmd.Parameters.Item(10).Value = tbasic.Text
    cmd.Execute()
    MsgBox(" Records Added Successfully")
End Sub

```

```
Private Sub ClearToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e
    As System.EventArgs) Handles ClearToolStripMenuItem.Click
```

```
    btupdate.Visible = False
```

```
    tempno.Text = ""
```

```
    tempname.Text = ""
```

```
    taddress.Text = ""
```

```
    tpho.Text = ""
```

```
    tdept.Text = ""
```

```
    tdob.Text = ""
```

```
    tdoj.Text = ""
```

```
    tbank.Text = ""
```

```
    tbasic.Text = ""
```

```
End Sub
```

```
Private Sub ExitToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles ExitToolStripMenuItem.Click
```

```
    End
```

```
End Sub
```

```
Private Sub FindToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e
    As System.EventArgs) Handles FindToolStripMenuItem.Click
```

```
    btupdate.Visible = False
```

```
    disp()
```

```
End Sub
```

```
Public Function disp()
```

```
    cmd.ActiveConnection = con
```

```
    no = Val(InputBox("Enter the Employee No "))
```

```
    cmd.CommandText = "select * from emp where empno=@empno"
```

```
    cmd.Parameters.Item(0).Value = no
```

```
    rs = cmd.Execute()
```

Do Until rs.EOF

```

tempno.Text = rs.Fields(0).Value
tempname.Text = rs.Fields(1).Value
taddress.Text = rs.Fields(2).Value
tpho.Text = rs.Fields(3).Value
tdesig.Text= rs.Fields(4).Value
tdept.Text = rs.Fields(5).Value
tdob.Text = rs.Fields(6).Value
tdoj.Text = rs.Fields(7).Value
comboxex.SelectedItem = rs.Fields(8).Value
tbank.Text = rs.Fields(9).Value
tbasic.Text = rs.Fields(10).Value
flag = 0
rs.MoveNext()

```

Loop

If flag = 1 Then

```

    MsgBox(" Record Not Found")

```

End If

End Function

```

Private Sub UpdateToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e
    As System.EventArgs) Handles UpdateToolStripMenuItem.Click

```

```

    disp()

```

```

    btupdate.Visible = True

```

End Sub

```

Private Sub FileToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles FileToolStripMenuItem.Click

```

End Sub

```

Private Sub btupdate_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles btupdate.Click
    cmd.ActiveConnection = con
    cmd.CommandText = "update emp set
        address=?,phone=?,desig=?,department=?,dob=?,doj=?,gender=?,baccno=?,basic=?
        where empno=?"
    cmd.Parameters.Item(0).Value = taddress.Text
    cmd.Parameters.Item(1).Value = tpho.Text
    cmd.Parameters.Item(2).Value = tdesig.Text
    cmd.Parameters.Item(3).Value = tdept.Text
    cmd.Parameters.Item(4).Value = tdob.Text
    cmd.Parameters.Item(5).Value = tdoj.Text
    cmd.Parameters.Item(6).Value = combosex.SelectedItem
    cmd.Parameters.Item(7).Value = tbank.Text
    cmd.Parameters.Item(8).Value = tbasic.Text
    cmd.Parameters.Item(9).Value = tempno.Text
    cmd.Execute()
    MsgBox(" Records Updated Successfully")
End Sub

```

```

Private Sub MonthlyPayToolStripMenuItem_Click(ByVal sender As System.Object,
    ByVal e As System.EventArgs) Handles MonthlyPayToolStripMenuItem.Click
    Me.Hide()
    monthpay.Show()
End Sub

```

```

Private Sub ReportToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e
    As System.EventArgs) Handles ReportToolStripMenuItem.Click
End Sub

```

```

Private Sub EmployeeReportToolStripMenuItem_Click(ByVal sender As
    System.Object, ByVal e As System.EventArgs) Handles
    EmployeeReportToolStripMenuItem.Click

    Me.Hide()

    Report.Show()

    Report.ReportViewer1.Show()

    Report.ReportViewer2.Hide()

End Sub

```

```

Private Sub PayReportToolStripMenuItem_Click(ByVal sender As System.Object,
    ByVal e As System.EventArgs) Handles PayReportToolStripMenuItem.Click

    Me.Hide()

    Report.Show()

    Report.ReportViewer1.Hide()

    Report.ReportViewer2.Show()

End Sub

End Class

```

Form design

The screenshot shows a Windows application window titled "monthpay". The menu bar contains "File" and "Return". The main form is titled "Payroll Processing". It contains the following elements:

- Employee No**: A text input field.
- DA**: A text input field.
- Employee Name**: A text input field.
- HRA**: A text input field.
- Designation**: A text input field.
- PF**: A text input field.
- Basic**: A text input field.
- CROSS**: A text input field.
- Net Pay**: A text input field followed by an **Edit** button.

Coding

```
Imports ADODB
```

```
Public Class monthpay
```

```
    Dim con As New ADODB.Connection
```

```
    Dim rs As New ADODB.Recordset
```

```
    Dim cmd1 As New ADODB.Command
```

```
    Dim flag As Integer = 1
```

```
    Dim no As Double
```

```
Private Sub ReturnToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e
    As System.EventArgs) Handles ReturnToolStripMenuItem.Click
```

```
    Me.Hide()
```

```
    combodesig.Show()
```

```
End Sub
```

```
Private Sub AddRecordToolStripMenuItem_Click(ByVal sender As System.Object,
    ByVal e As System.EventArgs) Handles AddRecordToolStripMenuItem.Click
```

```
    tempno.Enabled = True
```

```
    tempname.Enabled = True
```

```
    cmd1.ActiveConnection = con
```

```
    cmd1.CommandText = "insert into pay
        values(@Empno,@Empname,@desig,@basic,
        @hra,@da,@pf,@gpay,@netpay,@cmonth)"
```

```
    cmd1.Parameters.Item(0).Value = tempno.Text
```

```
    cmd1.Parameters.Item(1).Value = tempname.Text
```

```
    cmd1.Parameters.Item(2).Value = tdesig.Text
```

```
    cmd1.Parameters.Item(3).Value = tbasic.Text
```

```
    cmd1.Parameters.Item(4).Value = thra.Text
```

```
    cmd1.Parameters.Item(5).Value = tda.Text
```

```
    cmd1.Parameters.Item(6).Value = tpf.Text
```

```

cmd1.Parameters.Item(7).Value = tcross.Text
cmd1.Parameters.Item(8).Value = tnet.Text
cmd1.Parameters.Item(9).Value = Date.Now
cmd1.Execute()
MsgBox("Records Added Successfully ")
btedit.Visible = False
clr()
End Sub

```

```

Private Sub DeleteRecordToolStripMenuItem_Click(ByVal sender As System.Object,
        ByVal e As System.EventArgs) Handles DeleteRecordToolStripMenuItem.Click
    no = Val(InputBox("Enter the Employee No"))
    disp()
    tempno.Enabled = True
    tempname.Enabled = True
    If flag = 0 Then
        cmd1.CommandText = "delete * from pay where Empno=@empno"
        cmd1.Parameters.Item(0).Value = no
        cmd1.Execute()
        MsgBox(" Records Deleted Successfully")
        clr()
    End If
    btedit.Visible = False
End Sub

```

```

Private Sub EditRecordToolStripMenuItem_Click(ByVal sender As System.Object,
        ByVal e As System.EventArgs) Handles EditRecordToolStripMenuItem.Click
    no = Val(InputBox("Enter the Employee No"))
    disp()
    tempno.Enabled = False

```

```

tempname.Enabled = False
btedit.Visible = True
End Sub
Public Function disp()
    cmd1.ActiveConnection = con
    cmd1.CommandText = "select * from pay where Empno=@empno"
    cmd1.Parameters.Item(0).Value = no
    rs = cmd1.Execute()
    Do Until rs.EOF
        tempno.Text = rs.Fields(0).Value
        tempname.Text = rs.Fields(1).Value
        tdesig.Text = rs.Fields(2).Value
        tbasic.Text = rs.Fields(3).Value
        thra.Text = rs.Fields(4).Value
        tda.Text = rs.Fields(5).Value
        tpf.Text = rs.Fields(6).Value
        tcross.Text = rs.Fields(7).Value
        tnet.Text = rs.Fields(8).Value
        rs.MoveNext()
        flag = 0
    Loop
    If flag = 1 Then
        MsgBox(" Record Not Found")
    End If
End Function

Private Sub btedit_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles btedit.Click
    cmd1.ActiveConnection = con
    cmd1.CommandText = "update pay set
        desig=?,basic=?,hra=?,da=?,pf=?,gpay=?,netpay=? where empno=?"

```

```

cmd1.Parameters.Item(0).Value = tdesig.Text
cmd1.Parameters.Item(1).Value = tbasic.Text
cmd1.Parameters.Item(2).Value = thra.Text
cmd1.Parameters.Item(3).Value = tda.Text
cmd1.Parameters.Item(4).Value = tpf.Text
cmd1.Parameters.Item(5).Value = tcross.Text
cmd1.Parameters.Item(6).Value = tnet.Text
cmd1.Parameters.Item(7).Value = tempno.Text
cmd1.Execute()
MsgBox(" Records Edited Successfully")
btedit.Visible = False
End Sub

Private Sub monthpay_Load(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles MyBase.Load
con.Open("Provider=Microsoft.ACE.OLEDB.12.0;Data
    Source=C:\Users\dell\Desktop\rdbms\payroll\payroll\payroll.accdb")
tempno.Enabled = True
tempname.Enabled = True
btedit.Visible = False
tempname.Select()
End Sub

Public Function clr()
tempname.Text = ""
tempno.Text = ""
tdesig.Text = ""
tbasic.Text = ""
tda.Text = ""
thra.Text = ""
tpf.Text = ""

```

```

    tcross.Text = ""
    tnet.Text = ""
End Function

Private Sub tempno_LostFocus(ByVal sender As Object, ByVal e As
    System.EventArgs) Handles tempno.LostFocus
    cmd1.ActiveConnection = con
    cmd1.CommandText = "select * from emp where Empno=@empno"
    cmd1.Parameters.Item(0).Value = tempno.Text
    rs = cmd1.Execute()
    Do Until rs.EOF
        tempno.Text = rs.Fields(0).Value
        tempname.Text = rs.Fields(1).Value
        tdesig.Text = rs.Fields(2).Value
        tbasic.Text = rs.Fields(3).Value
        tda.Text = Val(tbasic.Text) * 0.1
        thra.Text = Val(tbasic.Text) * 0.05
        tpf.Text = Val(tbasic.Text) * 0.02
        tcross.Text = Val(tbasic.Text) + Val(tda.Text) + Val(thra.Text)
        tnet.Text = Val(tbasic.Text) + Val(tda.Text) + Val(thra.Text) - Val(tpf.Text)
        tempno.Enabled = True
        tempname.Enabled = True
        rs.MoveNext()
        flag = 0
    Loop
    If flag = 1 Then
        MsgBox(" Record Not Found")
    End If
End Sub
End Class

```

Output

Form1

File

Monthly Pay

Report

Exit

Payroll Processing

Employee No

223

Employee Name

Ravi

Date Of Birth

2/02/1990

Date Of Joining

5/03/2003

Address

KK Nagar,Chennai

Department Name

Tamil

Phone Number

98434424

Designation

Professor

Gender

Male

Basic

20000

Bank Account No

job4354546

monthpay

File

Return

Add Record

Delete Record

Edit Record

Payroll Processing

Employee No

111

DA

124345.3

Employee Name

ravi

HRA

62172.65

Designation

North street

PF

24869.06

Basic

1243453

CROSS

1429970.95

Net Pay

1405101.89

8. Aim: To Create a question and student database for conducting quiz and generate report for score details of the students

Table Name : quest

Field Name	Type	Description
Qno	Currency	Question Number
Question	Text	Question
Ans	Text	Answer for question
ch1	Text	Choice 1
ch2	Text	Choice 2
ch3	Text	Choice 3
ch4	Text	Choice 4

Table Name : student

Field Name	Type	Description
enrno	Currency	Enrollment Number
sname	Text	Student Name
cname	Text	Course Name
sex	Text	Sex
emailid	Text	emailID

Table Name : test

Field Name	Type	Description
enrno	Currency	Enrollment Number
sname	Text	Student Name
correct	Currency	No of correct
wrong	Currency	No of wrong
total	Currency	Total Marks

Form Design

The screenshot shows a Windows application window titled "Form1". The menu bar includes "File", "Question", "Test", "Report", and "Exit". A "File" menu is open, displaying "Add", "Delete", and a "Type Here" text box. The main content area is titled "Registration Form" and contains five text boxes for user input: "Enrollment No", "Name", "Course Name", "Gender", and "Email ID".

Coding

```
Imports ADODB
Public Class Form1
    Dim con As New ADODB.Connection
    Dim rs As New ADODB.Recordset
    Dim cmd As New ADODB.Command
    Dim flag As Integer = 1
    Dim no As String
    Private Sub Form1_Load(ByVal sender As System.Object, ByVal e As
        System.EventArgs) Handles MyBase.Load
        con.Open("Provider=Microsoft.ACE.OLEDB.12.0;Data
            Source=C:\Users\dell\Desktop\rdbms\quiz\quiz\quiz.accdb")
        MsgBox("Connected")
    EndSub
```

```
PrivateSub AddToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles AddToolStripMenuItem.Click
```

```
    cmd.ActiveConnection = con
    cmd.CommandText = "insert into student
        values(@Enrno,@sname,@cname,@sex,@emailid)"
    cmd.Parameters.Item(0).Value = tenrno.Text
    cmd.Parameters.Item(1).Value = tname.Text
    cmd.Parameters.Item(2).Value = tcname.Text
    cmd.Parameters.Item(3).Value = tgender.Text
    cmd.Parameters.Item(4).Value = temai.Text
    cmd.Execute()
    MsgBox(" Record Added Successfully")
    clr()
```

```
EndSub
```

```
PrivateSub DeleteToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles DeleteToolStripMenuItem.Click
```

```
    cmd.ActiveConnection = con
    no = InputBox("Enter the Student Name")
    cmd.CommandText = "select * from student where sname=@sname"
    cmd.Parameters.Item(0).Value = no
    rs = cmd.Execute()
```

```
DoUntil rs.EOF
```

```
    cmd.CommandText = "delete * from student where sname=@sname"
    cmd.Parameters.Item(0).Value = no
    rs = cmd.Execute()
    flag = 0
    MsgBox(" Record Deleted Successfully")
    clr()
```

```
Loop
```

```

If flag = 1 Then
    MsgBox(" InVaild Student Name")
EndIf
EndSub

PublicFunction clr()
    tenrno.Text = ""
    tcname.Text = ""
    tgender.Text = ""
    temai.Text = ""
EndFunction

PrivateSub TestToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles TestToolStripMenuItem.Click
    Me.Hide()
    test.Show()
EndSub

PrivateSub QuestionToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e
    As System.EventArgs) Handles QuestionToolStripMenuItem.Click
    Me.Hide()
    question.Show()
EndSub

PrivateSub ExitToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles ExitToolStripMenuItem.Click
    End
EndSub

```

```
PrivateSub ReportToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles ReportToolStripMenuItem.Click
```

```
Me.Hide()
```

```
Report.Show()
```

```
EndSub
```

```
EndClass
```

Question coding

The screenshot shows a Windows application window with the title bar 'question'. The main content area is titled 'Question Bank'. It contains several text input fields arranged vertically, each preceded by a label: 'Question Number', 'Question', 'Choice 1', 'Choice 2', 'Choice 3', 'Choice 4', and 'Answer'. At the bottom of the window, there are three buttons labeled 'Save', 'Delete', and 'Return'.

Question coding

```
PublicClassquestion
```

```
Dim con AsNew ADODB.Connection
```

```
Dim rs AsNew ADODB.Recordset
```

```

Dim cmd AsNew ADODB.Command
Dim flag AsInteger = 1
Dim no AsString
PrivateSub question_Load(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles MyBase.Load
    con.Open("Provider=Microsoft.ACE.OLEDB.12.0;Data
        Source=C:\Users\dell\Desktop\rdbms\quiz\quiz\quiz.accdb")
EndSub

```

```

PrivateSub btsave_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles btsave.Click
    cmd.ActiveConnection = con
    cmd.CommandText = "insert into quest
        values(@qno,@question,@ans,@ch1,@ch2,@ch3,@ch4) "
    cmd.Parameters.Item(0).Value = tqno.Text
    cmd.Parameters.Item(1).Value = tquest.Text
    cmd.Parameters.Item(2).Value = tans.Text
    cmd.Parameters.Item(3).Value = tch1.Text
    cmd.Parameters.Item(4).Value = tch2.Text
    cmd.Parameters.Item(5).Value = tch3.Text
    cmd.Parameters.Item(6).Value = tch4.Text
    cmd.Execute()
    MsgBox(" Question Added Successfully")
EndSub

```

```

PrivateSub btdelete_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles btdelete.Click
    cmd.ActiveConnection = con
Dim n AsInteger = Val(InputBox("Enter the question number"))
    cmd.CommandText = "select * from quest where qno=?"
    cmd.Parameters.Item(0).Value = n

```

```

        rs = cmd.Execute()
    DoUntil rs.EOF
        tqno.Text = rs.Fields(0).Value
        tquest.Text = rs.Fields(1).Value
        tans.Text = rs.Fields(2).Value
        tch1.Text = rs.Fields(3).Value
        tch2.Text = rs.Fields(4).Value
        tch3.Text = rs.Fields(5).Value
        tch4.Text = rs.Fields(6).Value
        cmd.CommandText = "delete * from quest where qno=?"
        cmd.Parameters.Item(0).Value = n
        cmd.Execute()
        rs.MoveNext()
        clr()
        flag = 0
        MsgBox(" Question Deleted Successfully")
    Loop
    If flag = 1 Then
        MsgBox(" Enter the Vaild Question Number")
    EndIf
EndSub
PublicFunction clr()
    tqno.Text = ""
    tquest.Text = ""
    tans.Text = ""
    tch1.Text = ""
    tch2.Text = ""
    tch3.Text = ""
    tch4.Text = ""
EndFunction

```

```

PrivateSub btreturn_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles btreturn.Click
    Me.Hide()
    Form1.Show()
EndSub
EndClass

```

Online test form

Online Test

Enrollment Number Name

☐ RadioButton1
 ☐ RadioButton2
 ☐ RadioButton3
 ☐ RadioButton4

QuestionNo	1	2	3	4

Coding

```

Public Classtest
Dim con As New ADODB.Connection
Dim rs As New ADODB.Recordset
Dim cmd As New ADODB.Command
Dim flag As Integer = 1
Dim no As String
Dim corr, wrng, t As Integer

```

```
PrivateSub bt1_Click(ByVal sender As System.Object, ByVal e As System.EventArgs)
    Handles bt1.Click, bt2.Click, bt3.Click, bt4.Click
```

```
    RadioButton1.Visible = True
```

```
    RadioButton2.Visible = True
```

```
    RadioButton3.Visible = True
```

```
    RadioButton4.Visible = True
```

```
    cmd.ActiveConnection = con
```

```
    cmd.CommandText = "select * from quest where qno=?"
```

```
    cmd.Parameters.Item(0).Value = sender.text
```

```
    rs = cmd.Execute()
```

```
DoUntil rs.EOF
```

```
    qno.Text = rs.Fields(0).Value
```

```
    quest.Text = rs.Fields(1).Value
```

```
    RadioButton1.Text = rs.Fields(3).Value
```

```
    RadioButton2.Text = rs.Fields(4).Value
```

```
    RadioButton3.Text = rs.Fields(5).Value
```

```
    RadioButton4.Text = rs.Fields(6).Value
```

```
    rs.MoveNext()
```

```
Loop
```

```
EndSub
```

```
PrivateSub HomeToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles HomeToolStripMenuItem.Click
```

```
Me.Hide()
```

```
Form1.Show()
```

```
EndSub
```

```
PrivateSub btsave_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs)
```

```
EndSub
```

```

PrivateSub test_Load(ByVal sender As System.Object, ByVal e As System.EventArgs)
    Handles MyBase.Load
    con.Open("Provider=Microsoft.ACE.OLEDB.12.0;Data
        Source=C:\Users\dell\Desktop\rdbms\quiz\quiz\quiz.accdb")
    Timer1.Start()
    t = 0
    corr = 0
    wrng = 0
    time.Text = 200
    RadioButton1.Visible = False
    RadioButton2.Visible = False
    RadioButton3.Visible = False
    RadioButton4.Visible = False
EndSub

PublicFunction disp()
    MsgBox(" Time Out")
    bt1.Enabled = False
    bt2.Enabled = False
    bt3.Enabled = False
    bt4.Enabled = False
EndFunction

PrivateSub Timer1_Tick(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles Timer1.Tick
    time.Text = Val(time.Text) - 1
    If Val(time.Text) <= 0 Then
        Timer1.Stop()
        disp()
    If corr = 3 Then

```

```

        wrng = 1
    ElseIf corr = 2 Then
        wrng = 2
    ElseIf corr = 1 Then
        wrng = 3
    Else
        wrng = 4
    EndIf
EndIf
EndSub

```

```

PrivateSub btfinish_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles btfinish.Click
    cmd.ActiveConnection = con
    cmd.CommandText = "insert into test
        values(@enrno,@sname,@correct,@wrong,@total)"
    cmd.Parameters.Item(0).Value = tenrno.Text
    cmd.Parameters.Item(1).Value = tname.Text
    cmd.Parameters.Item(2).Value = corr.ToString
    cmd.Parameters.Item(3).Value = wrng.ToString
    cmd.Parameters.Item(4).Value = (Val(corr.ToString) * 25)
    cmd.Execute()
    MsgBox("Marks Updated Successfully")
    Timer1.Stop()
EndSub

```

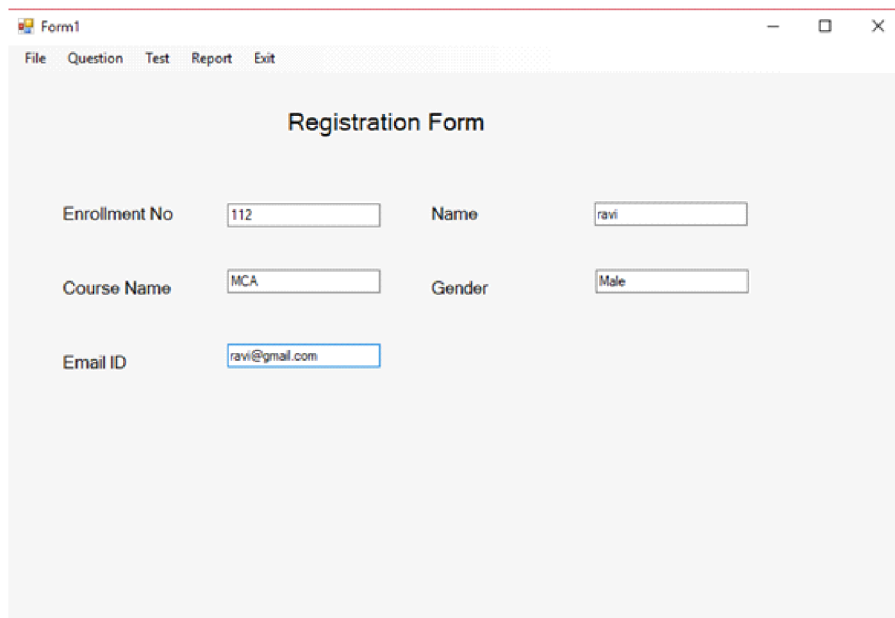
```

PrivateSub RadioButton1_Click(ByVal sender As Object, ByVal e As System.EventArgs)
    Handles RadioButton1.Click, RadioButton2.Click, RadioButton3.Click,
        RadioButton4.Click
    cmd.ActiveConnection = con
    cmd.CommandText = "select ans from quest where qno=?"

```

```
cmd.Parameters.Item(0).Value = qno.Text  
cmd.Execute()  
If qno.Text = 1 And RadioButton1.Checked = TrueThen  
    corr = corr + 1  
ElseIf qno.Text = 2 And RadioButton2.Checked = TrueThen  
    corr = corr + 1  
ElseIf qno.Text = 3 And RadioButton3.Checked = TrueThen  
    corr = corr + 1  
ElseIf qno.Text = 4 And RadioButton4.Checked = TrueThen  
    corr = corr + 1  
Else  
    wrng = wrng + 1  
EndIf  
EndSub  
EndClass
```

Output



The screenshot shows a Windows application window titled "Form1" with a menu bar containing "File", "Question", "Test", "Report", and "Exit". The main area of the window is titled "Registration Form". It contains five text input fields arranged in two columns. The first column has three fields: "Enrollment No" with the value "112", "Course Name" with the value "MCA", and "Email ID" with the value "ravi@gmail.com". The second column has two fields: "Name" with the value "ravi" and "Gender" with the value "Male".

Field	Value
Enrollment No	112
Course Name	MCA
Email ID	ravi@gmail.com
Name	ravi
Gender	Male

question

Question Bank

Question Number

Question

Choice 1

Choice 2

Choice 3

Choice 4

Answer

quiz

Question Added Successfully

Online Test

Home

Enrollment Number Name

0

Online Test

1 Who is the father of computer?

☐ Charles Babbage ☐ Tennis Ritchie

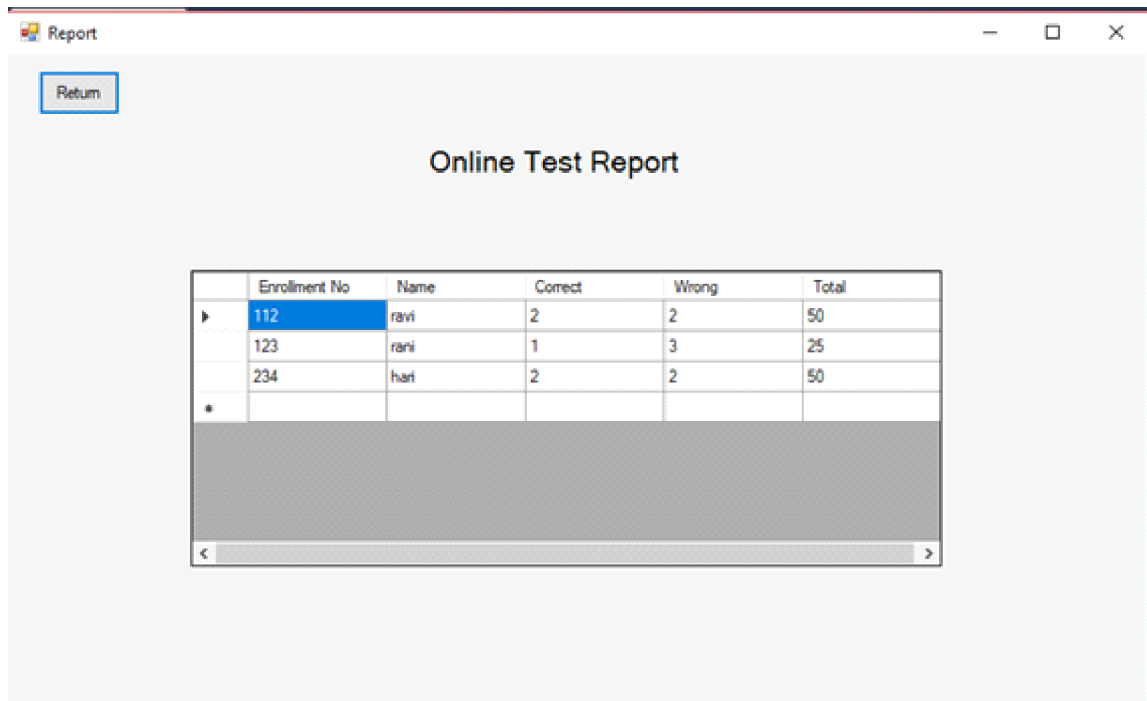
☐ Robert ☐ Bala Gurusamy

QuestionNo 1 2 3 4

quiz

Time Out

Report



	Enrollment No	Name	Correct	Wrong	Total
▶	112	ravi	2	2	50
	123	rani	1	3	25
	234	hari	2	2	50
*					

9. To create library master, member and transaction database for library maintenance and create reports for the same

Book table Name : master

Field Name	Type	Description
accno	Currency	Book Number
bname	Text	Name of the Book
aname	Text	Name of the Author
publ	Text	Name of the Publication
Price	Currency	Cost of the Book
cop	Currency	No of copies

Member Table : member

Field Name	Type	Description
enrno	Text	Enrollment Number
sname	Text	Name of the student
cname	Text	Name of the course
cocard	Currency	No of card

Issue return Table Name : trans

Field name	Type	Description
enrno	Text	Enrollment Number
sname	Text	Name of the student
cname	Text	Name of the course
bname	Text	Book Name
issuedt	Date	Issue date
redt	Date	Return date
nocard	Currency	No of cards
fine	Currency	Collect the fine

Form Designing:

Coding

```
Imports System.Data.OleDb

Public Class Form1

    Dim con As New OleDbConnection
    Dim cmd As New OleDbCommand
    Dim dr As OleDbDataReader

    Private Sub Form1_Load(ByVal sender As System.Object, ByVal e As
        System.EventArgs) Handles MyBase.Load

        con = New OleDbConnection("Provider=Microsoft.ACE.OLEDB.12.0;Data
            Source=C:\Users\dell\Desktop\rdbms\library\library\library.accdb")
        con.Open()
        combaccno.Visible = False
        btupdate.Visible = False
        MsgBox("Connected")
    EndSub
```

```
PrivateSub AddToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles AddToolStripMenuItem.Click
```

```
    combaccno.Visible = False
```

```
    taccno.Visible = True
```

```
    btupdate.Visible = False
```

```
    cmd.Connection = con
```

```
    cmd.CommandText = "insert into master
```

```
        values(@accno,@bname,@aname,@publ,@price,@cop)"
```

```
    cmd.Parameters.AddWithValue("@accno", Convert.ToInt32(taccno.Text))
```

```
    cmd.Parameters.AddWithValue("@bname", tbkname.Text)
```

```
    cmd.Parameters.AddWithValue("@aname", tauthor.Text)
```

```
    cmd.Parameters.AddWithValue("@publ", tpubl.Text)
```

```
    cmd.Parameters.AddWithValue("@price", Convert.ToDecimal(tprice.Text))
```

```
    cmd.Parameters.AddWithValue("@cop", Convert.ToInt32(tcopy.Text))
```

```
    cmd.ExecuteNonQuery()
```

```
    MsgBox("Records Added Successfully")
```

```
    taccno.Text = ""
```

```
    tbkname.Text = ""
```

```
    tauthor.Text = ""
```

```
    tpubl.Text = ""
```

```
    tprice.Text = ""
```

```
    tcopy.Text = ""
```

```
    con.Close()
```

```
EndSub
```

```
PrivateSub FindToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles FindToolStripMenuItem.Click
```

```
    combaccno.Visible = False
```

```
    taccno.Visible = True
```

```
    btupdate.Visible = False
```

```

Dim n, flag As Integer
    n = Val(InputBox("Enter the Book - Id "))
    cmd.Connection = con
    cmd.CommandText = "select * from master "
    dr = cmd.ExecuteReader()
While (dr.Read())
If (Convert.ToInt32(dr(0).ToString) = n) Then
    taccno.Text = dr(0).ToString
    tbkname.Text = dr(1).ToString
    tauthor.Text = dr(2).ToString
    tpubl.Text = dr(3).ToString
    tprice.Text = dr(4).ToString
    tcopy.Text = dr(5).ToString
    flag = 1
EndIf
EndWhile
If flag = 0 Then
    MsgBox("Not Found")
EndIf
    dr.Close()
EndSub

PrivateSub DeleteToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles DeleteToolStripMenuItem.Click
    combaccno.Visible = False
    taccno.Visible = True
    btupdate.Visible = False
Dim n, flag As Integer
    n = Val(InputBox("Enter the Book - Id "))
    cmd.Connection = con

```

```

        cmd.CommandText = "select * from master "
        dr = cmd.ExecuteReader()
    While (dr.Read())
        If (Convert.ToInt32(dr(0).ToString) = n) Then
            cmd = New OleDbCommand("delete from master where accno=@accno",
            con)
            cmd.Parameters.AddWithValue("@accno", n)
            cmd.ExecuteNonQuery()
            MsgBox("record deleted ")
            flag = 1
        EndIf
    EndWhile
    If flag = 0 Then
        MsgBox("Book Not Found")
    EndIf
EndSub

PrivateSub UpdateToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles UpdateToolStripMenuItem.Click
    combaccno.Visible = True
    Btupdate.Visible = True
    taccno.Visible = False
    cmd.Connection = con
    cmd.CommandText = "select * from master"
    dr = cmd.ExecuteReader()
    While dr.Read()
        combaccno.Items.Add(dr("accno").ToString)
    EndWhile
    dr.Close()
EndSub

```

PrivateSub btupdate_Click(ByVal sender As System.Object, ByVal e As
System.EventArgs) Handles btupdate.Click

Dim str As String

str = "Update master set bname=?,aname=?,publ=?,price=?,cop=? where
accno=?"

cmd = New OleDbCommand(str, con)

cmd.Parameters.AddWithValue("@bname", tbkname.Text)

cmd.Parameters.AddWithValue("@aname", tauthor.Text)

cmd.Parameters.AddWithValue("@publ", tpubl.Text)

cmd.Parameters.AddWithValue("@price", Convert.ToDecimal(tprice.Text))

cmd.Parameters.AddWithValue("@cop", Convert.ToInt32(tcopy.Text))

cmd.Parameters.AddWithValue("@accno", Convert.ToInt32(taccno.Text))

cmd.ExecuteNonQuery()

MsgBox("Records Updated Successfully")

taccno.Text = ""

tbkname.Text = ""

tauthor.Text = ""

tpubl.Text = ""

tprice.Text = ""

tcopy.Text = ""

combaccno.Visible = False

EndSub

PrivateSub combaccno_SelectedIndexChanged(ByVal sender As System.Object, ByVal
e As System.EventArgs) Handles combaccno.SelectedIndexChanged

cmd.Connection = con

cmd.CommandText = "select * from master where accno=@accno"

cmd.Parameters.AddWithValue("@accno", combaccno.SelectedItem)

dr = cmd.ExecuteReader()

While dr.Read()

```

        taccno.Text = dr(0).ToString
        tbkname.Text = dr(1).ToString
        tauthor.Text = dr(2).ToString
        tpubl.Text = dr(3).ToString
        tprice.Text = dr(4).ToString
        tcopy.Text = dr(5).ToString
    EndWhile
EndSub

PrivateSub ExitToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles ExitToolStripMenuItem.Click
End
EndSub

PrivateSub TransactionToolStripMenuItem_Click(ByVal sender As System.Object, ByVal
    e As System.EventArgs) Handles TransactionToolStripMenuItem.Click
Me.Hide()
Transcation.Show()
EndSub

PrivateSub BookReportToolStripMenuItem_Click(ByVal sender As System.Object, ByVal
    e As System.EventArgs) Handles BookReportToolStripMenuItem.Click
Me.Hide()
report.Show()
    report.ReportViewer1.Show()
    report.ReportViewer2.Hide()
EndSub

PrivateSub IssuseReportToolStripMenuItem_Click(ByVal sender As System.Object,
    ByVal e As System.EventArgs) Handles IssuseReportToolStripMenuItem.Click
Me.Hide()

```

```

report.Show()
    report.ReportViewer2.Show()
    report.ReportViewer1.Hide()
EndSub
EndClass

```

Transaction form

Transaction

Member File Issue/Return Home Type Here

Add Issue
Find Return
Delete Type Here
Type Here

Library Maintenance System

Enrollment Number Student Name Course Name

Book ID Book Title No of Cards

Issue Date Return Date Fine Amount

	Book ID	Book Name	Publisher	No Of Copies
*				

Return

Coding

```

Imports System.Data.OleDb

Public Class Transaction
    Dim con As New OleDbConnection
    Dim cmd, cmd1, cmd2 As New OleDbCommand
    Dim dr, dr1 As OleDbDataReader
    Dim enno, flag, cop As Integer
    Dim bname As String
    Private Sub Transaction_Load(ByVal sender As System.Object, ByVal e As
        System.EventArgs) Handles MyBase.Load

```

'TODO: This line of code loads data into the 'LibraryDataSet1.master' table. You can move, or remove it, as needed.

```
Me.MasterTableAdapter.Fill(Me.LibraryDataSet1.master)
```

'TODO: This line of code loads data into the 'LibraryDataSet2.master' table. You can move, or remove it, as needed.

'TODO: This line of code loads data into the 'LibraryDataSet1.member' table. You can move, or remove it, as needed.

```
con = New OleDbConnection("Provider=Microsoft.ACE.OLEDB.12.0;Data
    Source=C:\Users\dell\Desktop\rdbms\library\library\library.accdb")
```

```
con.Open()
```

```
btissue.Visible = False
```

```
bt.Visible = False
```

```
btret.Visible = False
```

```
EndSub
```

```
PrivateSub AddToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles AddToolStripMenuItem.Click
```

```
bt.Visible = True
```

```
btissue.Visible = False
```

```
btret.Visible = False
```

```
taccno.Enabled = False
```

```
tbktitle.Enabled = False
```

```
tissue.Enabled = False
```

```
tretdt.Enabled = False
```

```
tfine.Enabled = False
```

```
EndSub
```

```
PrivateSub HomeToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles HomeToolStripMenuItem.Click
```

```
Me.Hide()
```

```
Form1.Show()
```

```
EndSub
```

```
PrivateSub MemberFileToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles MemberFileToolStripMenuItem.Click
```

```
    taccno.Enabled = False
```

```
    tbktitle.Enabled = False
```

```
    tissue.Visible = False
```

```
    tretdt.Visible = False
```

```
    tfine.Enabled = False
```

```
EndSub
```

```
PrivateSub DeleteToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles DeleteToolStripMenuItem.Click
```

```
    bt.Visible = False
```

```
    btissue.Visible = False
```

```
    btret.Visible = False
```

```
Dim n, flag As Integer
```

```
    n = Val(InputBox("Enter the Member - Id "))
```

```
    cmd.Connection = con
```

```
    cmd.CommandText = "select * from member "
```

```
    dr = cmd.ExecuteReader()
```

```
While (dr.Read())
```

```
If (Convert.ToInt32(dr(0).ToString) = n) Then
```

```
    cmd = New OleDbCommand("delete from member where enrno=@enrno", con)
```

```
    cmd.Parameters.AddWithValue("@enrno", n)
```

```
    cmd.ExecuteNonQuery()
```

```

        MsgBox("Member deleted ")
        flag = 1
    EndIf
EndWhile
If flag = 0 Then
    MsgBox("Member Not Found")
EndIf
    dr.Close()
EndSub

PrivateSub FindToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles FindToolStripMenuItem.Click
    bt.Visible = False
    btissue.Visible = False
    btret.Visible = False
    Dim n, flag As Integer
    n = Val(InputBox("Enter the Book - Id "))
    cmd.Connection = con
    cmd.CommandText = "select * from member "
    dr = cmd.ExecuteReader()
    While (dr.Read())
        If (Convert.ToInt32(dr(0).ToString) = n) Then
            terno.Text = dr(0).ToString
            tname.Text = dr(1).ToString
            tcourse.Text = dr(2).ToString
            tnocard.Text = dr(3).ToString
            flag = 1
        EndIf
        terno.Text = ""
        tname.Text = ""
    EndWhile
EndSub

```

```

        tcourse.Text = ""
        tnocard.Text = ""
    EndWhile
    If flag = 0 Then
        MsgBox("Not Found")
    EndIf
    dr.Close()
EndSub

PrivateSub IssueToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles IssueToolStripMenuItem.Click
    btissue.Visible = True
    bt.Visible = False
    btret.Visible = False
    cop = 0
    Dim n, flag As Integer
    n = Val(InputBox("Enter the Book ID"))
    enrno = Val(InputBox("Enter the Student Enrollment No"))
    cmd1.Connection = con
    cmd1.CommandText = "select * from master where accno=@accno"
    cmd1.Parameters.AddWithValue("@accno", Val(n))
    dr = cmd1.ExecuteReader()
    While (dr.Read())
    If dr(5) > 0 Then
        cop = dr(5).ToString
        MsgBox(cop)
        flag = 1
        taccno.Text = n
        tbktitle.Text = dr(1).ToString
        tissue.Text = Date.Now
    EndIf
    EndWhile
EndSub

```

```

        trettdt.Text = Convert.ToDateTime(tissue.Text).AddDays(15).ToString
        tfine.Text = 0
    Else
        MsgBox(" Book Not Available")
    EndIf
EndWhile
    dr.Close()
    dispmember()
EndSub

PrivateSub btissue_Click(ByVal sender As Object, ByVal e As System.EventArgs)
    Handles btissue.Click
    cmd = New OleDbCommand("update master set cop=? where accno=?", con)
    cmd.Parameters.AddWithValue("@cop", cop - 1)
    cmd.Parameters.AddWithValue("@accno", taccno.Text)
    cmd.ExecuteNonQuery()
    cmd.Dispose()
    'update no of card
    cmd1.Connection = con
    cmd1 = New OleDbCommand("update member set nocard=? where enrno=?",
        con)
    cmd1.Parameters.AddWithValue("@nocard", Val(tnocard.Text) - 1)
    cmd1.Parameters.AddWithValue("@enrno", terno.Text)
    cmd1.ExecuteNonQuery()
    cmd2.CommandText = "insert into trans
        values(@enrno,@sname,@cname,@bname,@issuedt,@redt,@nocard,@fine)"
    cmd2.Connection = con
    cmd2.Parameters.AddWithValue("@enrno", terno.Text)
    cmd2.Parameters.AddWithValue("@sname", tname.Text)
    cmd2.Parameters.AddWithValue("@cname", tcourse.Text)

```

```

cmd2.Parameters.AddWithValue("@bname", tbktitle.Text)
cmd2.Parameters.AddWithValue("@issuedt", tissue.Text)
cmd2.Parameters.AddWithValue("@redt", tretdt.Text)
cmd2.Parameters.AddWithValue("@nocard", tnocard.Text)
cmd2.Parameters.AddWithValue("@fine", 0)
cmd2.ExecuteNonQuery()
MsgBox("Book Issused")
flag = 1
btissue.Enabled = False
bt.Enabled = False
btret.Enabled = False
dr.Close()
If flag = 0 Then
    MsgBox("Book not Issuse")
EndIf
EndSub

PrivateSub ReturnToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles ReturnToolStripMenuItem.Click
    bt.Visible = False
    btissue.Visible = False
    btret.Visible = True
    cop = 0
    bname = InputBox("Enter the Book Name ")
    enrno = Val(InputBox("Enter the Student ENrollment No "))
    cmd.Connection = con
    cmd.CommandText = "select * from trans where enrno=@enrno and
        bname=@bname"
    cmd.Parameters.AddWithValue("@enrno", enrno)
    cmd.Parameters.AddWithValue("@bname", bname)

```

```

        dr1 = cmd.ExecuteReader()
While (dr1.Read())
    terno.Text = dr1(0).ToString
    tname.Text = dr1(1).ToString
    tcourse.Text = dr1(2).ToString
    tbktitle.Text = dr1(3).ToString
    tissue.Text = dr1(4).ToString
    tretdt.Text = dr1(5).ToString
    tnocard.Text = dr1(6).ToString
    flag = 0
EndWhile
If Date.Compare(CDate(tretdt.Text), CDate(Date.Now)) < 0 Then
    tfine.Text = 100
Else
    tfine.Text = 0
EndIf
dr1.Close()
EndSub

PrivateSub btret_Click(ByVal sender As System.Object, ByVal e As System.EventArgs)
    Handles btret.Click
If flag = 1 Then
    MsgBox(" Invalid Book ID")
Else
    cmd1.Connection = con
    cmd1.CommandText = "select * from member where enrno=@enrno "
    cmd1.Parameters.AddWithValue("@enrno", enrno)
    cmd1.ExecuteNonQuery()
    dr1 = cmd1.ExecuteReader()
While (dr1.Read())

```

```

        cmd1 = NewOleDbCommand("update member set nocard=? where
enrno=?", con)
        cmd1.Parameters.AddWithValue("@nocard", Val(tnocard.Text) + 1)
        cmd1.Parameters.AddWithValue("@enrno", dr1(0).ToString)
        cmd1.ExecuteNonQuery()
    EndWhile
    dr1.Close()
    cmd.Connection = con
    cmd.CommandText = "select * from master "
    dr1 = cmd.ExecuteReader()
    While (dr1.Read())
    If (dr1(1).ToString = bname) Then
        flag = 1
        tbktitle.Text = dr1(1).ToString
        taccno.Text = dr1(0).ToString
        cmd = NewOleDbCommand("update master set cop=? where accno=?",
con)
        cmd.Parameters.AddWithValue("@cop", Convert.ToInt32(dr1(5).ToString) +
1)
        cmd.Parameters.AddWithValue("@accno", dr1(0).ToString)
        cmd.ExecuteNonQuery()
        cmd.Dispose()
        cmd2 = NewOleDbCommand("update trans set redt=?,fine=?,nocard=?
where enrno=? and bname=?", con)
        cmd2.Parameters.AddWithValue("@redt", tretdt.Text)
        cmd2.Parameters.AddWithValue("@fine", tfine.Text)
        cmd2.Parameters.AddWithValue("@nocard", Val(tnocard.Text) + 1)
        cmd2.Parameters.AddWithValue("@enrno", terno.Text)
        cmd2.Parameters.AddWithValue("@bname", tbktitle.Text)

```

```

        cmd2.ExecuteNonQuery()
        MsgBox("Book Return")
    EndIf
EndWhile
If flag = 0 Then
    MsgBox("Book not Return")
EndIf
EndIf
EndSub
PublicFunction dispmember()
    cmd.Connection = con
    cmd.CommandText = "select sname,cname,nocard from member where
        enrno=?"
    cmd.Parameters.AddWithValue("@enrno", enrno.ToString)
    dr = cmd.ExecuteReader()
    While dr.Read()
        tname.Text = dr(0).ToString
        tcourse.Text = dr(1).ToString
        tnocard.Text = dr(2).ToString
        terno.Text = enrno
    EndWhile
    dr.Close()
EndFunction

PrivateSub bt_Click(ByVal sender As System.Object, ByVal e As System.EventArgs)
    Handles bt.Click
    cmd.Connection = con
    cmd.CommandText = "insert into member
        values(@enrno,@sname,@cname,@nocard)"
    cmd.Parameters.AddWithValue("@enrno", terno.Text)

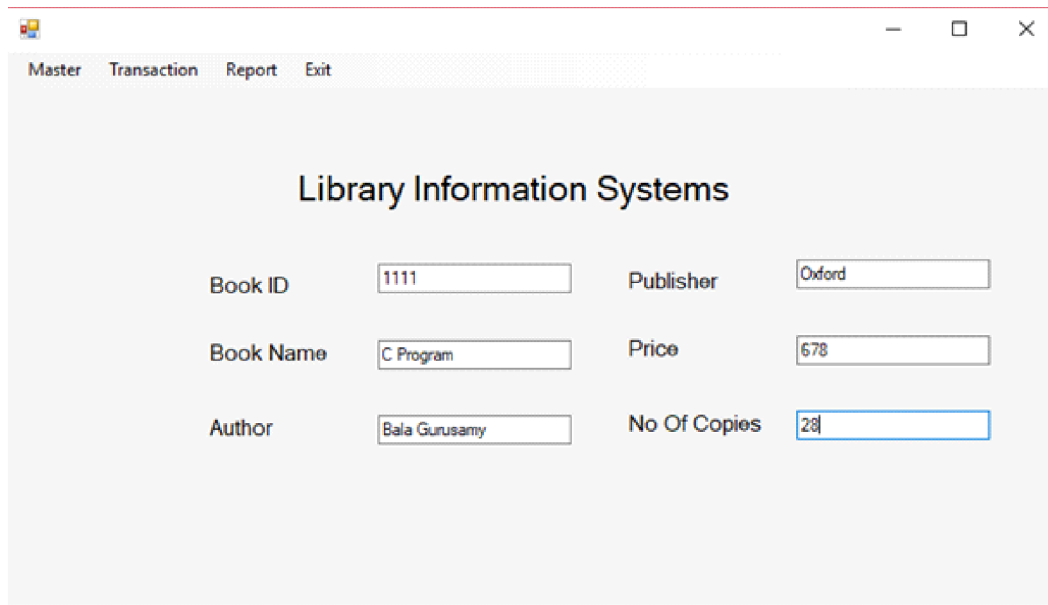
```

```

cmd.Parameters.AddWithValue("@sname", tname.Text)
cmd.Parameters.AddWithValue("@cname", tcourse.Text)
cmd.Parameters.AddWithValue("@nocard", Convert.ToInt32(tnocard.Text))
cmd.ExecuteNonQuery()
MsgBox("Member Added Successfully")
terno.Text = ""
tname.Text = ""
tcourse.Text = ""
tnocard.Text = ""
taccno.Enabled = True
tbktitle.Enabled = True
tissue.Visible = True
trettdt.Visible = True
tfine.Visible = True
EndSub

PrivateSub tnocard_LostFocus(ByVal sender As Object, ByVal e As System.EventArgs)
    Handles tnocard.LostFocus
    If Val(tnocard.Text < 0) Then
        MsgBox("Card Not Available")
        btissue.Enabled = False
    EndIf
EndSub
EndClass

```

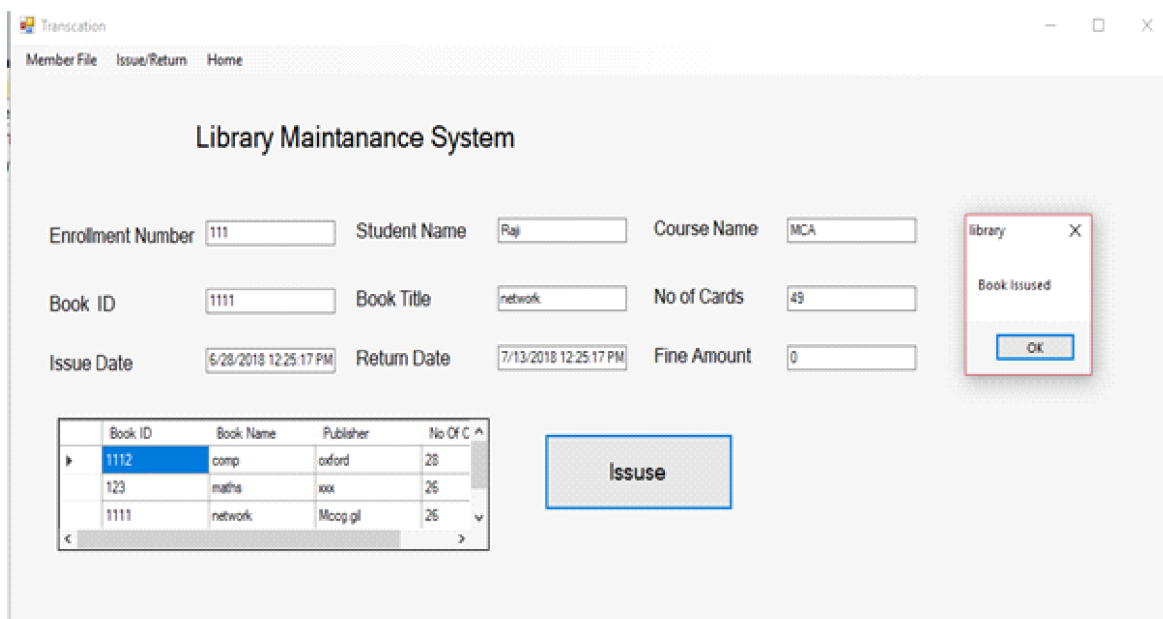
Output

Library Information Systems

Book ID: 1111 Publisher: Oxford

Book Name: C Program Price: 678

Author: Bala Gurusamy No Of Copies: 28



Library Maintenance System

Enrollment Number: 111 Student Name: Raj Course Name: MCA

Book ID: 1111 Book Title: network No of Cards: 49

Issue Date: 6/28/2018 12:25:17 PM Return Date: 7/13/2018 12:25:17 PM Fine Amount: 0

Book ID	Book Name	Publisher	No Of C
1112	comp	oxford	28
123	maths	xxx	26
1111	network	Mccg gil	26

Issue

library Book Issused OK

Form2

Home

Book ID	Book Name	Publisher	Price	No of Copies
1111	C Program	oxford	678	28
1112	Data Structure	ISBN	400	26
1113	network	Mccg.gil	556	25

Form2

Home

Enrollment No	Name	Book Name	Issusedate	Return date	No of Card	Fine
1111	Raja	Maths	12/5/2018 12:00:00 AM	12/15/2018 12:00:00 AM	2	100
1113	Hari	Network	6/21/2018 6:41:18 PM	7/6/2018 6:41:18 PM	34	0.0000

10. Aim: To create table for Telephone directory and provide the option to search and display.

Table structure

Field Name	Type	Description
custno	Currency	Customer Number
custname	Text	Name of the Customer
address	Text	Name of the address
phoneno	Currency	Phone Number

Form Design

Coding

```
Imports System.Data.OleDb
Public Class Form1
    Dim con As New OleDbConnection
    Dim cmd, cmd1 As New OleDbCommand
    Dim dr As OleDbDataReader
    Dim n As Double
    Dim flag As Integer = 0
```

```

PrivateSub Form1_Load(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles MyBase.Load
    con = New OleDbConnection("Provider=Microsoft.ACE.OLEDB.12.0;Data
        Source=C:\Users\dell\Desktop\rdbms\telephone\telephone\telephone.accdb")
    con.Open()
    combcustno.Visible = False
    btupdate.Visible = False
    MsgBox("Connected")
EndSub

```

```

PrivateSub AddToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles AddToolStripMenuItem.Click
    combcustno.Visible = False
    tcustno.Visible = True
    btupdate.Visible = False
    cmd.Connection = con
    cmd.CommandText = "insert into tele values(@custno,@custname,
        @address,@phoneno)"
    cmd.Parameters.AddWithValue("@custno", Convert.ToInt32(tcustno.Text))
    cmd.Parameters.AddWithValue("@custname", tcustname.Text)
    cmd.Parameters.AddWithValue("@address", taddress.Text)
    cmd.Parameters.AddWithValue("@phoneno",
        Convert.ToDouble(tphoneno.Text))
    cmd.ExecuteNonQuery()
    MsgBox("Record Added")
    dr.Close()
EndSub

```

```

PrivateSub ExitToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles ExitToolStripMenuItem.Click

```

End

EndSub

```
PrivateSub ClearToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles ClearToolStripMenuItem.Click
```

```
    combcustno.Visible = False
```

```
    tcustno.Visible = True
```

```
    tcustno.Text = ""
```

```
    tcustname.Text = ""
```

```
    taddress.Text = ""
```

```
    tphoneno.Text = ""
```

EndSub

```
PrivateSub DeleteToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles DeleteToolStripMenuItem.Click
```

```
    combcustno.Visible = False
```

```
    tcustno.Visible = True
```

```
    btupdate.Visible = False
```

```
    n = Val(InputBox("Enter the customer Number"))
```

```
    cmd.Connection = con
```

```
    cmd.CommandText = "select custno from tele"
```

```
    dr = cmd.ExecuteReader()
```

```
    cmd1.Connection = con
```

```
While (dr.Read())
```

```
If (Convert.ToInt32(dr(0)) = n) Then
```

```
    cmd1.CommandText = "delete * from tele where custno=@custno"
```

```
    cmd1.Parameters.AddWithValue("@custno", n)
```

```
    cmd1.ExecuteNonQuery()
```

```
    flag = 1
```

```
    MsgBox("Record Deleted")
```

EndIf

EndWhile

If flag = 0 Then

 MsgBox("Invalid Customer Number")

EndIf

 dr.Close()

EndSub

PrivateSub EditToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles EditToolStripMenuItem.Click

 combcustno.Visible = True

 tcustno.Visible = False

 btupdate.Visible = True

 cmd.Connection = con

 cmd.CommandText = "select custno from tele"

 dr = cmd.ExecuteReader()

While (dr.Read())

 combcustno.Items.Add(dr(0).ToString)

EndWhile

 dr.Close()

EndSub

PrivateSub btupdate_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles btupdate.Click

 cmd = New OleDbCommand("update tele set custname=?,address=?,phoneno=?
 where custno=?")

 cmd.Connection = con

 cmd.Parameters.AddWithValue("@custname", Convert.ToInt32(tcustno.Text))

 cmd.Parameters.AddWithValue("@address", taddress.Text)

```

cmd.Parameters.AddWithValue("@phoneno", Convert.ToDouble(tphoneno.Text))
cmd.Parameters.AddWithValue("@custno", combcustno.SelectedItem)
cmd.ExecuteNonQuery()
MsgBox("Record updated")
dr.Close()
EndSub

PrivateSub combcustno_SelectedIndexChanged(ByVal sender As System.Object,
    ByVal e As System.EventArgs) Handles combcustno.SelectedIndexChanged
EndSub

PrivateSub SearchToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles SearchToolStripMenuItem.Click
Me.Hide()
search.Show()
EndSub

PrivateSub ReportToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As
    System.EventArgs) Handles ReportToolStripMenuItem.Click
Me.Hide()
Report.Show()
EndSub

PrivateSub taddress_LostFocus(ByVal sender As Object, ByVal e As
    System.EventArgs) Handles taddress.LostFocus
cmd.Connection = con
cmd.CommandText = "select max(phoneno) from tele"
dr = cmd.ExecuteReader()
While dr.Read
    tphoneno.Text = Val(dr(0).ToString) + 1

```

```

EndWhile
    MsgBox(" Mobile No Generated")
    dr.Close()
EndSub

PrivateSub combcustno_SelectedValueChanged(ByVal sender As Object, ByVal e As
    System.EventArgs) Handles combcustno.SelectedValueChanged
    cmd.Connection = con
    cmd.CommandText = "select * from tele where custno=@custno"
    cmd.Parameters.AddWithValue("@custno",
        Convert.ToInt32(combcustno.SelectedItem))
    dr = cmd.ExecuteReader()
    While dr.Read()
        tcustno.Text = dr(0).ToString
        tcustname.Text = dr(1).ToString
        taddress.Text = dr(2).ToString
        tphoneno.Text = dr(3).ToString
    EndWhile
    dr.Close()
EndSub
EndClass
Search form design

```

Coding

```

Imports System.Data.OleDb
Public Class search
    Dim con As New OleDbConnection
    Dim n As Double
    Dim flag As Integer = 0

    PrivateSub search_Load(ByVal sender As System.Object, ByVal e As

```

```

        System.EventArgs) Handles MyBase.Load
con = New OleDbConnection("Provider=Microsoft.ACE.OLEDB.12.0;Data
        Source=C:\Users\dell\Desktop\rdbms\telephone\telephone\telephone.accdb")
con.Open()
Dim dr1 As OleDbDataReader
Dim cmd As New OleDbCommand
        cmd.Connection = con
        cmd.CommandText = "select custname from tele "
        dr1 = cmd.ExecuteReader()
While (dr1.Read())
        combcustname.Items.Add(dr1(0).ToString)
EndWhile
        cmd.Dispose()
        dr1.Close()
EndSub

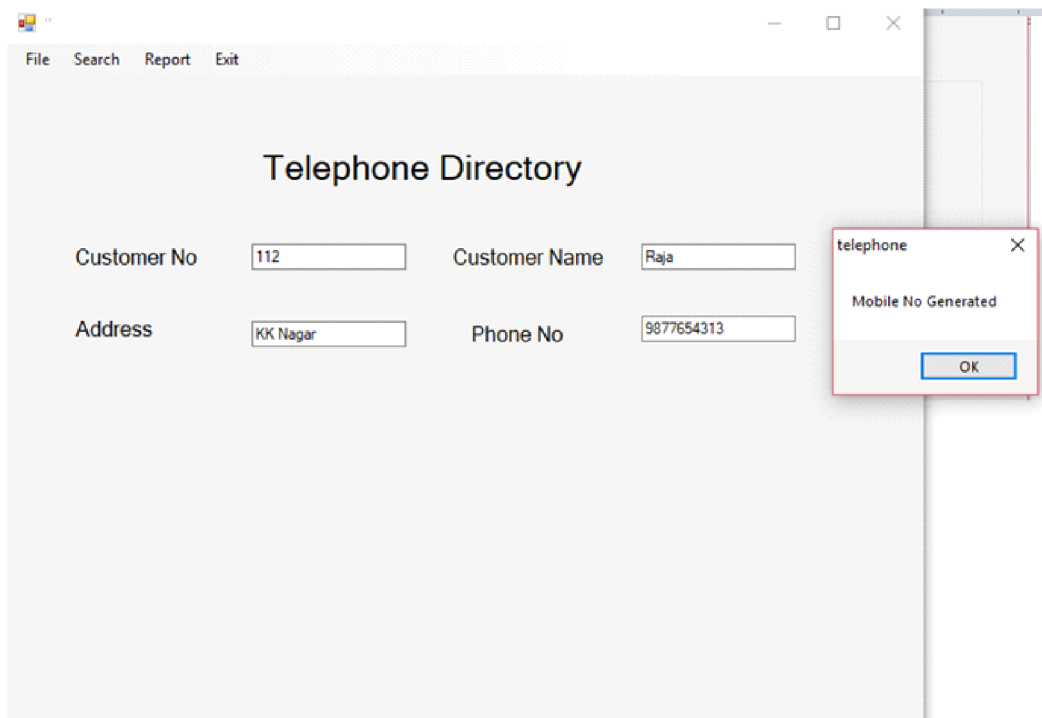
PrivateSub HomeToolStripMenuItem1_Click(ByVal sender As System.Object, ByVal e As
        System.EventArgs) Handles HomeToolStripMenuItem1.Click
Me.Hide()
Form1.Show()
EndSub

PrivateSub combcustname_SelectedIndexChanged(ByVal sender As System.Object,
        ByVal e As System.EventArgs) Handles
        combcustname.SelectedIndexChanged
Dim dr As OleDbDataReader
Dim cmd As New OleDbCommand
        cmd.Connection = con
cmd.CommandText = "select address,phoneno from tele where
        custname=@custname"

```

```
cmd.Parameters.AddWithValue("@custname", combcustname.SelectedItem.ToString)
dr = cmd.ExecuteReader()
While (dr.Read())
    taddress.Text = dr(0).ToString
    tpho.Text = dr(1).ToString
EndWhile
cmd.Dispose()
dr.Close()
EndSub
EndClass
```

Output



search

Home

S.S.N.L Telephone Directory Maintance

Search By Name

Customer Name

Phone No

Address

Report

Report

Home

Telephone Directory

	Customer No	Name	Address	Phone No
▶	112	Raja	KK Nagar	9877654312
	113	selvi	Ashok nagar	8976655444
	114	Hari	T nagar	8906554455
*				